ITU ARI-Bulletin of the Istanbul Technical University

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Aerodynamic and Heat Transfer Research Facilities for Modern Aero-Propulsion Units with Special Emphasis on Rotating Turbine Stages

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Abstract: This paper presents state-of-the-art experimental aero-heat-transfer studies conducted in rotating turbine research facilities. Turbine heat-transfer research has made significant progress over the last few decades. Since the full-scale operational conditions of a modern gas turbine dictate high temperatures well in excess of 2000°C (3600 °F) and pressure ratios ranging from 20 to 50, experimental forced-convection heat-transfer research on the gas side of a rotating turbine environment is technically challenging. This paper provides a brief review of turbine heat-transfer research across various facilities, including short-duration blowdown and large-scale/low-speed turbine systems. Because the final outcome of any forced-convection heat-transfer problem depends on the detailed structure of momentum transfer in highly 3D, unsteady, rotating, and turbulent viscous flows, emphasis will also be placed on pertinent turbine aerodynamic features. The most significant parameters to simulate in a rotating aero-heat-transfer facility include the Reynolds number based on the blade chord, the Mach number for compressibility and shock-wave effects, tip speed, the intensity and scale of free-stream turbulence, the Strouhal number for unsteady wake passing effects, the free-stream-to-wall temperature ratio, the coolant-to-free-stream temperature ratio, the specific heat ratio, the molecular Prandtl number of the operating gas, and a rotation number (Rosby number) for turbine aero-heat-transfer work performed under rotational conditions. A flow coefficient and a loading coefficient defined by the actual turbine hardware are typically maintained during laboratory testing.

Keywords: Aero-propulsion, axial turbines, aerodynamics, heat transfer, advanced experimentation

Modern Havacılık İtki Sistemleri için Aerodinamik ve Isı Transferi Araştırma Tesisleri, Özellikle Dönel Türbin Kademelerine Odaklanarak

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Özet: Bu makale, döner türbin araştırma tesislerinde yürütülen en son deneysel aero-ısı transferi çalışmalarını sunmaktadır. Türbin ısı transferi araştırmaları son birkaç on yılda önemli ilerleme kaydetmiştir. Modern bir gaz türbininin tam ölçekli çalışma koşulları, 2000°C (3600°F) 'nin çok üzerinde yüksek sıcaklıklar ve 20 ila 50 arasında değişen basınç oranları gerektirdiğinden, döner türbin ortamının gaz tarafında deneysel taşınım ısı transferi araştırması teknik olarak zor bir görevdir. Bu makale, kısa süreli/yuksek hızlı ve büyük ölçekli/düşük hızlı test sistemleri de dahil olmak üzere çeşitli tesislerdeki türbin ısı transferi araştırmalarına ilişkin sınırlı bir inceleme sunmaktadır. Herhangi bir taşınım ısı transferi probleminin karakteri, 3 boyutlu, kararsız, dönel ve türbülanslı sürtünmeli akış ortamlarında momentum transferinin ayrıntılı yapısıyla yakından ilişkili olduğundan, ilgili türbin aerodinamik özelliklerine de vurgu yapılmaktadır. Döner bir aero-ısı transfer tesisinde simüle edilecek en önemli parametreler, kanat boyutuna ile tanimlanan Reynolds sayısı, sıkıştırılabilirlik ve şok dalgası etkileri için Mach sayısı, kanat ucu hızı, serbest akış türbülansının derecesi ve ölçeği, kararsız iz geçiş etkileri için Strouhal sayısı, serbest akış/duvar sıcaklık oranı, soğutucu/serbest akış sıcaklık oranı, özgül ısı oranı, çalışma gazının moleküler Prandtl sayısı ve dönme koşulları altında gerçekleştirilen türbin aero-ısı transfer calismalari için bir dönüş sayısı (Rosby number) olarak sıralanabilir. Laboratuvar testleri sırasında, gerçek türbin donanımı tarafından tanımlanan bir akış katsayısı ve bir yükleme katsayısı genellikle korunur.

Anahtar Kelimeler: havacilik itki sistemleri, eksenel turbinler,aerodinamik, isi transferi, ileri deneysel yontemler

REVIEW/RESEARCH PAPER

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Citation: Camci,C. (2026), Contemporary Aero-Heat Transfer Research Infrastructure With Special Emphasis on Rotating Facilities, ITU ARI Bulletin of the İstanbul Technical University 56,(2) 1–12

Submission Date : 7 February 2026
Online Acceptance : 15 February 2026
Online Publishing : 15 February 2026

1.Introduction

The experimental rotating research rigs for gas turbine aero-heat transfer studies range from full-scale gas turbine demonstrators to short-duration facilities, from rotating coolant passage simulators to large-scale/low-speed turbine research facilities, from rotating disk cavity research rigs to tip leakage flow simulators. While instrumented full-scale gas turbine demonstrators are excellent candidates for generating highly realistic gas turbine heat-transfer data, the initial investment required to construct them, their extremely high operational costs, and the technological challenges of performing reliable, high-resolution aero-thermal measurements limit their current use.

Short-duration facilities, or blow-down-type rigs, have been extremely popular choices mainly because of their relatively low operating costs and the excellent scaling available for Mach and Reynolds number sensitive aero-thermal phenomena. Although they may not operate at full-scale gas temperature, they can still generate measurable heat transfer

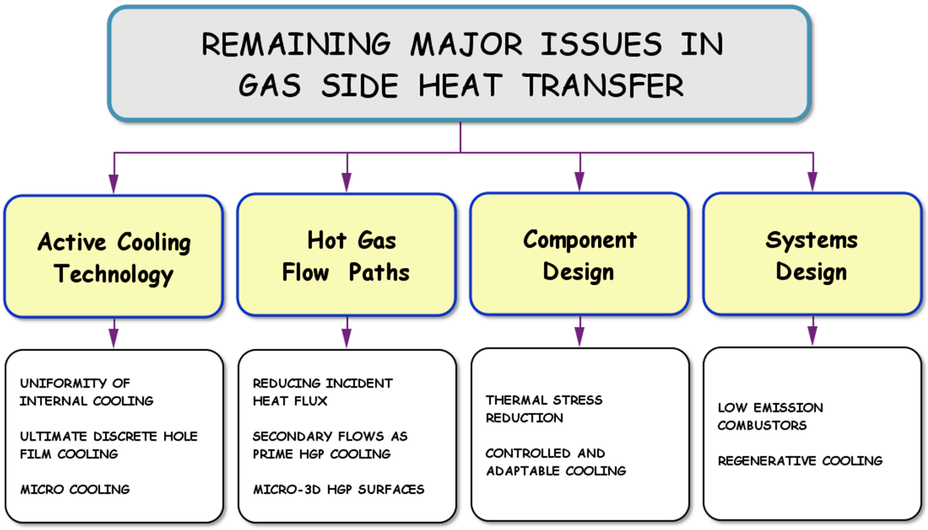


Figure 1. Major gas side heat transfer issues in turbines, Bunker [2006]

from the gas side to turbine blades in a linear cascade, annular cascade, or turbine-stage arrangement, where rotor operation is usually realistic. The heat transfer coefficients obtained from short-duration rotating rigs are considered reasonably reliable, given the experimental uncertainties. Although heat transfer instrumentation in short-duration facilities requires advanced heat flux sensor fabrication, signal transmission, and signal processing capabilities, the results generated in these facilities have extremely high technical value during and after a turbine design sequence. The most visible aero-heat transfer facilities were constructed and operated by Oxford University, Von Karman Institute for Fluid Dynamics, Ohio State University (Calspan previously), MIT Gas Turbine Lab, Wright Patterson Air Force Base, QinetiQ Corporation UK, Texas A&M, DLR Germany, ETH Zurich, Notre Dame University, Zucrow/PETAL lab at Purdue, Whittle Lab UK and Penn State University.

Large-scale, low-speed research facilities are also frequently used in aero-heat transfer research because of their low operating costs and high reliability and operational safety. One of the first low-speed turbine research rigs was operated by the United Technologies Research Center in Connecticut. The currently operational low-speed, large-scale turbine research rigs are located at the Aero-Heat Transfer Laboratory at Pennsylvania State University, Purdue University, and Texas A&M University. There are also a number of low-speed/large-scale turbine research facilities currently operational in Germany and Switzerland, mainly used for the improvement of stage aerodynamic features.

This paper provides an overview of currently operational turbine heat-transfer research facilities with a fully operational rotor. In addition to major rotating rigs, the research results from the Axial Flow Turbine Research Facility AFTRF at Penn State, operated by the author of this paper, will be presented.

Since the final status of any forced-convection heat-transfer problem is determined by the detailed structure of momentum transfer in highly unsteady, rotating, and turbulent viscous-flow environments, emphasis will be placed on pertinent turbine aerodynamic features in rotating facilities.

2. Current Gas Side Heat Transfer Issues

The ten major issues in gas turbine aerothermal research that remain today were presented by Bunker [2006]. He mentioned that effective solutions to these ten issues are required to address the remaining gas turbine problems in realizable engineering. Figure 1 shows a general classification of the challenging problems. The present-day gas turbine heat transfer technology typically meets the strict requirements of modern gas turbine engines. The improvements in this technology in the areas of efficiency and quality of life came in small increments over a long period. The most significant achievements on today's technological plateau were the widespread use of shaped film-cooling holes and the implementation of thermal barrier coatings. Recent advances over the last few years have focused on understanding detailed flow physics and on developing and using advanced analytical/computational tools for aero-thermal problems in the gas side of turbine engines. Bunker [2006] considers several important questions that must be answered to further aero-thermal technology. His first question is whether to cool a component only at selected hot spots or it globally. His second question is whether the overall shielding performance against incident heat load can be improved beyond the level achieved by TBCs. He considers maintaining a component temperature closer to the hot-gas temperature as one way to minimize heat loads from the gas side. A more isothermal component is likely to have a smaller internal thermal stress field. Most modern turbine systems are effectively cooled, and they usually operate well above the maximum allowable material temperature. High heat loads in modern systems, along with effective cooling, lead to greater through-wall and in-plane thermal gradients. Moving thermal energy to another location and dealing with it there is also considered by Bunker [2006]. A comprehensive treatment of convective heat transfer and aerodynamics in axial flow turbines can be found in an IGTI Gas Turbine Scholar Lecture by Dunn [2001].

3. Research Simulation Issues

Proper scaling of the gas turbine operating environment requires replicating the Reynolds number, the molecular Prandtl number of the gases, the Mach number, the wall-to-free-stream temperature ratio, and the specific heat ratio of the gases. The simulation parameters listed here can be obtained directly from the non-dimensional forms of the momentum conservation and thermal energy equations.

and the specific heat ratio. This condition is in addition to Re and Pr, "for proper simulation ability" in a forced convection heat transfer problem. Simulation of a gas turbine environment also needs to include a proper mass-flow function, corrected speed, and a rotation number, also called the Rossby number. The last set of requirements directly results from a dimensional analysis of the stagnation enthalpy, power output, and efficiency of a turbomachinery system. The scaling issues considered so far have assumed a laminar-flow environment in a momentum and convective heat-exchange problem. In a modern gas turbine, the combustor exit flow exhibits significant turbulence and free-stream unsteadiness. The turbulence intensity and length scale of turbulence are the two parameters that should be considered very carefully. The rotor blades of a gas turbine continuously pass through the highly turbulent, low-axial-momentum wakes of nozzle guidevanes, generating a highly unsteady flow in the rotor

$$\rho^* \frac{\partial V_i^*}{\partial t^*} + \rho^* V_j^* \frac{\partial V_i^*}{\partial x_j^*} = -\frac{\partial p^*}{\partial x_i} + \frac{1}{\mathrm{Re}} \frac{\partial^2 V_i^*}{\partial x_j^* \partial x_j^*} \quad \text{where} \quad \mathrm{Re} = \frac{\rho_o V_o L}{\mu_o} = \frac{V_o L}{\nu_o} \tag{1}$$

$$\rho^* c_p^* \frac{DT^*}{Dt^*} = \frac{V_o^2}{(T_w - T_o) c_{p_o}} \frac{Dp^*}{Dt^*} + \frac{k_o}{V_o \rho_o c_{p_o}} \nabla^2 T^* + \frac{\mu_o V_o}{L \rho_o c_{p_o} (T_w - T_o)} \Phi^* \quad \text{where} \quad \Phi^* = \frac{\Phi}{\mu_o V^2 / L^2} \tag{2}$$

Viscous dissipation rate in its dimensional form is defined as follows:

$$\Phi = \mu \cdot \left[2\left(\frac{\partial u}{\partial x}\right)^2 + 2\left(\frac{\partial v}{\partial y}\right)^2 + 2\left(\frac{\partial w}{\partial z}\right)^2 + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}\right)^2 + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right)^2 \right] + \lambda \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right)^2 \tag{3}$$

The final form of the energy equation, as shown in equation 4, indicates that for a proper simulation of energy transfer in any environment where forced-convection heat transfer is significant, one needs to simulate the Reynolds number Re, the Prandtl number Pr, and the Eckert number Ec.

$$\rho^* c_p^* \frac{DT^*}{Dt^*} = Ec \cdot \frac{Dp^*}{Dt^*} + \frac{1}{\mathrm{Re} \cdot \mathrm{Pr}} \nabla^2 T^* + \frac{Ec}{\mathrm{Re}} \cdot \Phi^* \quad \text{where} \quad Ec = \frac{V_o^2}{(T_w - T_o) c_{p_o}} \quad \text{and} \quad \mathrm{Pr} = \frac{\mu_o c_{p_o}}{k_o} \tag{4}$$

The Eckert number represents the ratio of the kinetic energy of the mean flow to the thermal energy-carrying capacity of the molecular domain. Eckert number may be easily substituted with "free-stream gas to wall temperature ratio", Reynolds number and Mach number in the region of interest. The equation of state $p_\infty = \rho R T_\infty$ and the local speed of sound $a = \sqrt{\gamma R T_\infty}$ can be rearranged to show $a^2 = \gamma . p_\infty / \rho_\infty$ = $c_{p_o}(\gamma - 1). T_\infty$. After re-arrangement, the Eckert number can be expressed as follows :

passages. In nozzle guide vanes with a transonic/supersonic exit flow, the rotor blades may also chop through shock waves originating from the NGV trailing edges. In the relative flow environment of the rotor exposed to hot gases, the blade surfaces experience the periodic passage of NGV wakes, shock waves, and high-momentum fluid originating from the central sections of the upstream passage. The wall shear stress generation on gas turbine surfaces and heat transfer to these surfaces are highly three-dimensional and unsteady. When the gas turbine is film-cooled, the typical additional

$$Ec = \frac{\dfrac{V_o^2}{a^2}}{\dfrac{(T_w - T_o) c_{p_o}}{c_{p_o} (\gamma - 1) \cdot T_\infty}} = (\gamma - 1) \cdot M^2 \cdot \frac{T_\infty}{(T_w - T_o)} = (\gamma - 1) \cdot M^2 \cdot \frac{\dfrac{T_\infty}{T_o}}{(\dfrac{T_w}{T_o} - 1)} \quad \text{where} \quad \frac{T_o}{T_\infty} = \left[1 + (\gamma - 1) \cdot M^2\right] \tag{5}$$

The final form of the energy equation suggests that one needs to simulate the Eckert number by operating at a realistic Mach number M, the free-stream-to-wall temperature ratio $T_{o\infty}/T_w$,

scaling parameters are the blowing rate, which is the coolant-to-free-stream mass flux ratio $M=\rho_c U_c/\rho_\infty U_\infty$, and the coolant-to-free-stream temperature ratio, T_c/T_o. The ratio of the coolant to free-stream momentum flux rate is also a popular cooling performance parameter $I=\rho_c U_c^2/\rho_\infty U_\infty^2$. A mass flow rate ratio of coolant to free-stream for a stage or the entire turbine may also be used as a cooling performance indicator.

4. Key Simulation Parameters

The most important scaling parameter in a gas turbine heat-transfer problem is the free-stream gas-to-wall temperature ratio. The very high gas temperatures in a modern turbine make full-scale investigations technically challenging and expensive. Figure 2 shows the combustor exit temperature and pressure for several advanced gas turbine configurations that have been successfully operated in the past. Instead of working with the absolute level of the free-stream gas temperature, operating with the "free-stream gas to wall temperature ratio" would allow the momentum and heat transfer in a gas turbine to be effectively simulated under relatively manageable testing conditions. In general, reducing the absolute temperature has a significant advantage in the design of a turbine test facility. Lowered free-stream gas temperatures require a lower operating pressure when the Reynolds number is fully matched between the actual operation and the test facility.

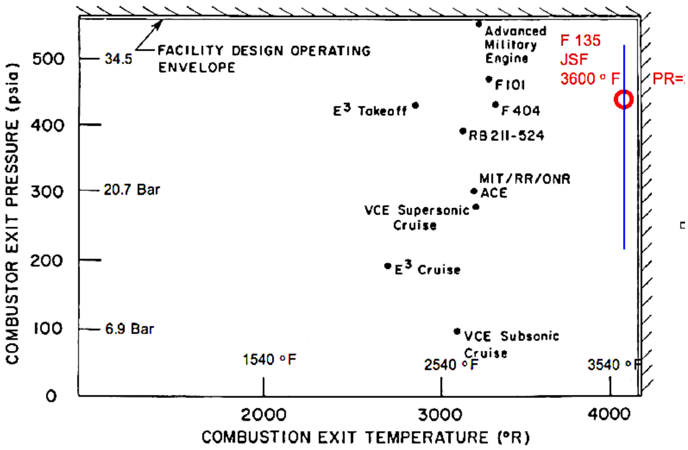


Figure 2. Typical gas turbine entry temperatures and pressures from existing gas turbine systems

Lowering the gas temperature during operation has another significant benefit. The rotational speed of the test facility also decreases as the absolute gas temperature decreases. The mechanical stresses on the test facility and the rotating instrumentation system are reduced simultaneously, simplifying facility design and operation. The reduction in rotational speed reduces the frequency response requirements of instrumentation, especially for time-accurate and phase-locked measurements in the rotor frame of reference, Epstein et al. [1985]. The measurement of unsteady heat transfer, wall shear stress, aspirating probe measurements for simultaneous total pressure and temperature measurements, hot wire measurements, hot film measurements, time accurate tip clearance measurements, particle image velocimetry measurements and pressure sensitive paint measurements all fall into a category that benefits from relatively reduced frequency response operation at lowered rotational speed in facilities realistically simulating present day turbines.

Although the low-speed operation of a research turbine may not be sensitive to the specific heat ratio $\gamma = c_p/c_v$, a proper matching of this fluid property can not be ignored for transonic and supersonic turbines. The specific heat ratio γ is a strong property of the gas composition that can be adjusted by using a combination of low and high γ molecules. Increasing the molecular weight of the gases in a turbine reduces the required operating pressure at a given Reynolds number and lowers the gas's speed of sound, thereby reducing the turbine's rotational speed. The heaviest mono-atomic gas which is readily available is Argon, Epstein et al., [1985]. For example, a 75 % Argon and 25 % Freon-12 mixture is an ideal composition for scaled turbine testing. Freon-14 may be an effective candidate for turbine cooling, but its use is currently restricted for environmental reasons. The turbine disk's rotational speed is usually adjusted to keep the corrected speed constant. Scaled turbine operation requires an operation at a significantly reduced mass flow rate and power level.

5. Aero-Heat Transfer Research in Simulated Turbine Rigs

The rotating turbine research rigs for aero-heat transfer studies range from full-scale gas turbine demonstrators to short-duration facilities, from rotating coolant passage simulators to large-scale/low-speed turbine research facilities, from rotating disk cavity research rigs to tip leakage flow simulators. While instrumented full-scale gas turbine demonstrators are excellent facilities for generating highly realistic gas turbine heat-transfer data and for performance evaluation, the initial investment required to construct them, their extremely high operational costs, and the technological challenges of performing reliable, high-resolution aero-thermal measurements limit their current use.

The short-duration facilities have been popular choices because of their relatively low operating costs and excellent scalability for Mach- and Reynolds-number-sensitive aero-thermal phenomena. Although they may not operate at the full-scale free-stream gas temperature, they can still generate measurable heat transfer from the gas side to the turbine blades. The heat transfer coefficients obtained from short-duration rotating rigs are considered high quality, given the experimental uncertainties. Although heat transfer instrumentation in short-duration facilities requires advanced heat flux sensor fabrication, signal transmission, and signal processing capabilities, the results generated in these facilities have extremely high technical value during and after a turbine design sequence. The most visible short-duration heat-transfer facilities were constructed and operated by Oxford University, the Von Karman Institute for Fluid Dynamics, Ohio State University (previously Calspan), MIT Gas Turbine Lab, Wright-Patterson Air Force Base Laboratories, and QinetiQ Corporation.

Large-scale, low-speed research facilities are also frequently used in aero-heat transfer research because of their low operating costs, high reliability, and operational safety. One of the first turbine research rigs was operated at the United Technologies Research Center in Connecticut. The currently operational low-speed, large-scale turbine research rigs are located at the Aero-Heat Transfer Laboratory at Pennsylvania State University, Purdue University, and Texas A&M University. There are also several low-speed/large-scale turbine research facilities currently operational, used mainly to improve stage aerodynamic features.

6. A Comparative Evaluation of Turbine Rigs

A Proposed Blow Down Turbine/Fan Test Facility at UTRC: A blow-down research facility was designed by UTRC for aero-heat transfer studies in turbines and fans, Case, [1985]. Although this facility was never built, the design had a unique capability to test both a turbine and a fan in a short-duration mode. Figure 3 shows the turbine model and fan model connected between a common supply tank and a large dump tank. The turbine test section was designed to include a multi-stage, cooled turbine, an adjustable throttle valve, a rotor spin-up motor, and an eddy-current brake for torque absorption and speed control. The facility allowed up to 4 turbine stages, with a maximum diameter of 37 inches. The quick-acting inlet valve, which operates in less than 50 ms and is located inside the supply tank, was one of the most important components in this turbine test rig, enabling a total test duration of 1 second. This 1 second test duration included the initial transient disturbances caused by the rapid opening of the inlet valve. The facility inlet included provisions to simulate inlet pressure and temperature profiles, and a heat exchanger section for turbine inlet temperature control. A 100 HP spin-up motor located in the initially evacuated dump tank brought the turbine rotor to its operating speed before the quick-acting inlet valve opened. The pressure ratio between the supply tank and the dump tank controlled the Mach number in this facility's simulation. The supply tank level, the supply tank initial pressure, and the throttle valve control the mass flow rate and Reynolds number in the turbine rig. The facility design allowed testing every 20 minutes and had provisions for boundary layer bleed at the turbine inlet.

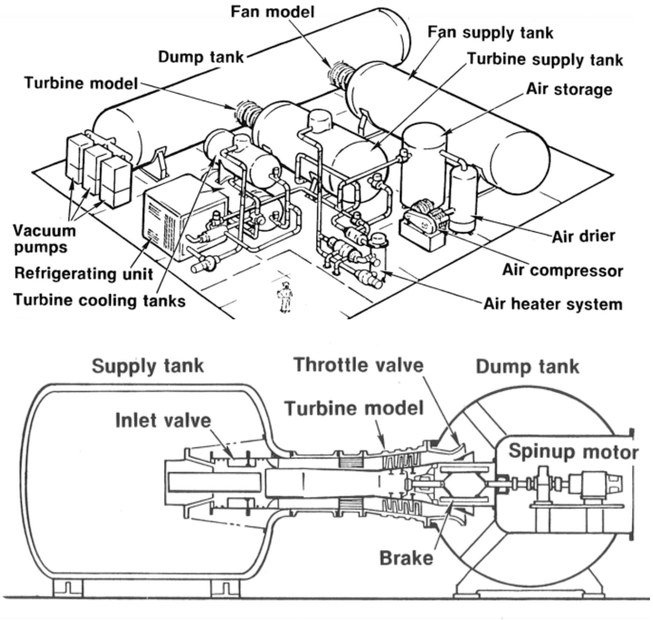


Figure 3. Proposed turbine/fan test facility at UTRC, Case [1985] (not constructed)

MIT Blowdown Turbine Test Facility: -The MIT turbine test rig is a short-duration facility that is capable of testing film-cooled, high-work aircraft engine stages for a typical duration of 400 ms, Epstein et al. [1984]. This facility can generate test-rig conditions that accurately simulate turbine operation up to an inlet pressure of 40 atm and a turbine inlet temperature of 4000 ° F (2500 ° K). A 0.5 m-diameter test facility is intended to investigate three-dimensional, unsteady turbine flow and convective heat transfer. A full experimental simulation of a turbine stage, including wall to free-stream temperature ratio, coolant to free-stream temperature ratio, Reynolds number, Mach number, Rossby number, corrected speed, weight flow and specific heat ratio, is possible in a relatively large diameter stage in which detailed unsteady flow, heat transfer and performance indicating instrumentation is available. Figure 4 shows major turbine design parameters and turbine scaling characteristics, as well as a mid-span blade-to-blade view of

TABLE 1
TURBINE DESIGN PARAMETERS

Turbine loading, $\Delta H/U^2$	-2.3
Total pressure ratio	4.2
Velocity ratio, C_x/U	0.63
Rotor aspect ratio	1.5
NGV exit Mach No.	1.18

TABLE 2
MIT BLOWDOWN TURBINE SCALING

	Full Scale	MIT Blowdown
Fluid	Air	Ar-Fr 12
Ratio specific heats	1.28	1.28
Mean metal temperature	1118°K (1550°F)	295°K (72°F)
Metal/gas temp. ratio	0.63	0.63
Inlet total temperature	1780°K (2750°F)	478°K (400°F)
True NGV chord	8.0 cm	5.9 cm
Reynolds number*	2.7×10^6	2.7×10^6
Inlet pressure, atm	19.6	4.3
Outlet pressure, atm	4.5	1.0
Outlet total temperature	1280°K (1844°F)	343°K (160°F)
Prandtl number	0.752	0.755
Eckert number†	1.0	1.0
Rotor speed, rpm	12,734	6,190
Mass flow, kg/sec	49.0	16.6
Power, watts	24,880,000	1,078,000
Test time	Continuous	0.3 sec

* Based on NGV chord and isentropic exit conditions
† $(\gamma-1)M^2T/\Delta T$

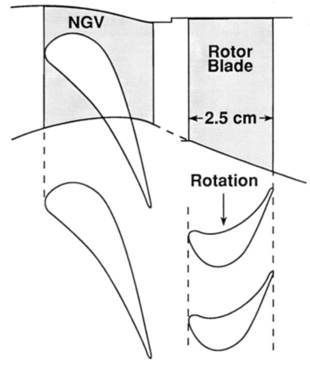


Fig. 1: Transonic turbine geometry investigated

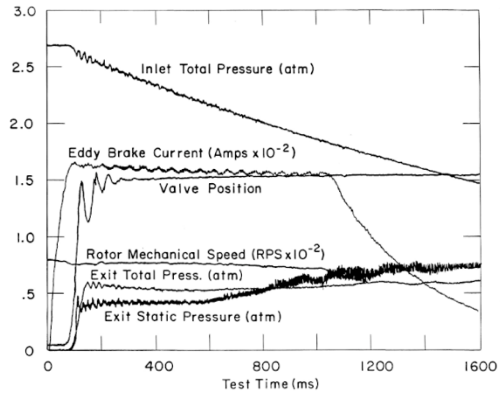


Figure 4. MIT blow down rig design parameters, turbine scaling, and timing, Epstein et al. [1984]

the turbine stage. Epstein et al. [1984] mention that instrumentation access, initial cost, and high time-response requirements are the primary concerns in the design of this facility. The simulation characteristics shown in Figure 4 indicate that a realistic free-stream gas-to-wall temperature ratio of 0.63 can be achieved at 295 ° K for full-scale operation at 1780 ° K (2750 ° F). A full-scale coolant temperature of 790 ˚K is reduced to 212 ˚K in the test rig. The lowered free-stream gas temperature in the rig allows a turbine disk operation at almost half speed (N=6190 rpm) compared with full-scale operation (12,734 rpm). The MIT blowdown turbine facility operates at a turbine inlet total pressure of 4.3 atm (64 psia), whereas the full-scale hardware runs at 19.6 atm (289 psia). Other important test rig parameters, compared with the full-scale hardware, are shown in Figure 4. The specific heat ratio in the facility is adjusted using a combination of a monoatomic gas, Argon, and other heavier molecular-weight gases in refrigerant gas mixtures. The fast-opening valve isolating the test section from the supply tank opens fully in 50 ms and is designed for a supply tank pressure of 10 atm when the test section is near vacuum conditions. The supply tank temperature can be adjusted to as high as 530 °K. A 0.9 m-diameter annular cone-shaped plug seals against the inner and outer annuli of the flow path, as shown in Figure 4. The cone slides forward, opening graphite rings along its shaft to form a smooth annular contraction from the supply tank to the test section. The contraction is part of the boundary layer control system, which also includes a boundary layer bleed arrangement. Time-resolved turbine rotor-blade heat-transfer measurements performed in the MIT blowdown turbine rig are compared with computations by Abhari et al. [1991]. The The influence of hardware manufacturing tolerances on variations in rotor heat transfer is discussed. A multi-layer thin-film heat-flux gauge with a 25 μμm-thick polyimide insulator is adhesively bonded to the blade profile. The sensing area is

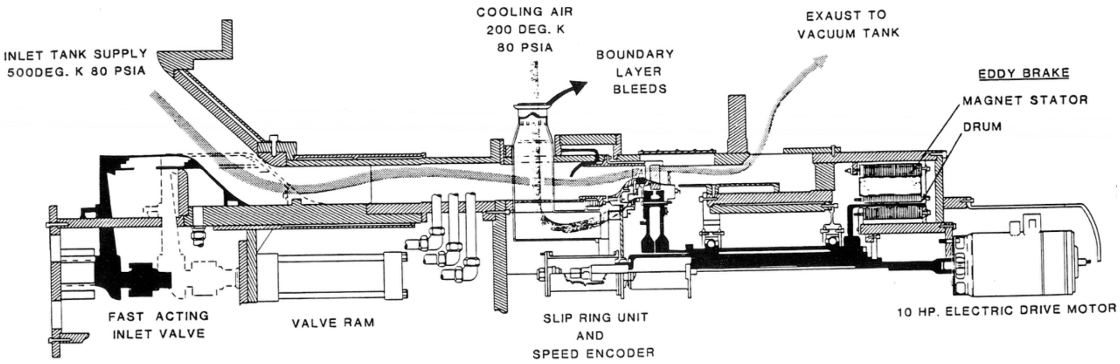
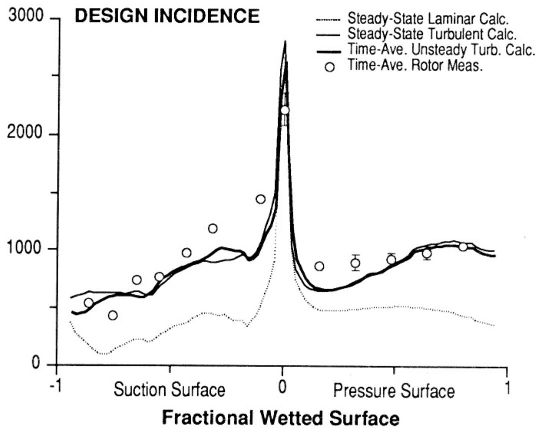
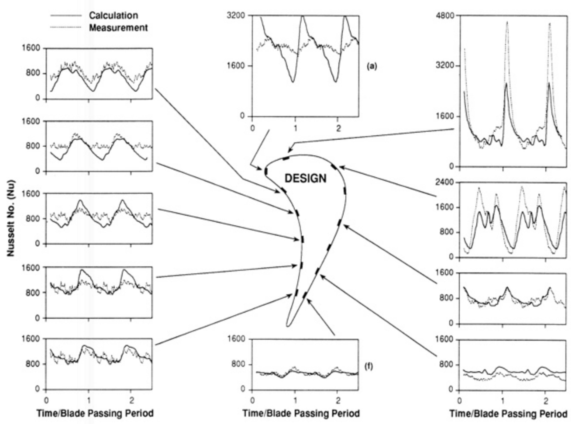


Figure 5. MIT turbine test section and fast-acting inlet valve, Epstein et al. [1984]

limited to 1 x 1.3 mm. Above 1 kHz, the 25 μm-thick insulator appears infinitely thick to the top surface.The entire frequency domain of heat flux measurements, from d.c. to 100 kHz, is reconstructed, with the blade-passing frequency at 3.6 kHz. The heat flux signals are digitized at a 100 kHz sampling rate and ensemble-averaged over 360 vane-passing periods. The absolute calibration accuracy of the time-accurate heat flux sensors is quoted to be about 10 %, Abhari et al [1991].





Fig

turbine facility, Abhari et al. [1991]

A comparison of the time-averaged measured rotor heat transfer, the unsteady NGV-rotor calculation, and the steady-state solution for the rotor alone is shown in Figure 6 for design incidence. The Nusselt number is defined using the blade axial chord, heat flux, inlet relative total minus local bladetemperature and thermal conductivity at the blade local temperature. The difference between steady-state turbulent and time-averaged unsteady calculations is minimal. The measured heat transfer rates generally follow the fully turbulent computational levels. Overall, the predicted heat transfer rate is about 11% lower than the measured value. Time-resolved heat transfer data at design incidence are presented in Figure 6 along both the suction and pressure surfaces. The influence of unsteady flow on heat-transfer signals is shown. Near the stagnation point, computations using a Baldwin-Lomax turbulence model show much higher unsteadiness than the measurements. The unsteadiness in heat-transfer data results from NGV wake impingement on the rotor blade surfaces and potential coupling, including moving shock patterns emanating from the NGV trailing edges. A very interesting conclusion from Abhari et al.'s [1991] study was that the prediction of heat transfer in highly loaded turbines is more sensitive to the details of the inviscid flow prediction than to the particulars of the turbulence model employed. Another interesting observation in his study was the small difference between the steady, fully turbulent rotor calculation and the time-averaged unsteady heat-transfer calculation. The unsteadiness played a very small role in the specific highly loaded transonic turbine, even though both the measurements and computations show very significant unsteady fluctuations in time-accurate heat transfer data.

UTRC's Large-Scale Rotating Rig (LSRR): United Technologies Large-Scale Rotating Rig (LSSR) was one of the first rotating turbine research facilities that performed a significant amount of aerodynamics and heat transfer research in a low-speed turbine environment, where the first three rows of the test rig represented a high-bypass ratio aircraft engine. General characteristics of this rotating test facility is described in Dring and Joslyn [1981]. This 1 ½-stage low-speed facility operated under laboratory air conditions at an inlet total pressure of about 1 atm and an inlet total temperature of 294 º K, with a flow coefficient of 0.78. The experimental turbine, with a 0.8 hub-to-tip ratio, had 22 first vanes, 28 first blades, and 28 second vanes. The model airfoil chord lengths were about five times the full-scale hardware chord lengths. The rotor speed was about 410 m/s, and the airfoil Reynolds number was approximately 560,000, typical of high-pressure turbine airfoils. The effects of turbulence and stator-rotor interactions on turbine heat transfer were studiedby Blair et al. [1988 a,b]. Figure 8 presents the Stanton number distribution on the suction side of the LSSR rotor blade in comparison to Sharma et al.'s [1990] computations. A very good match between the unsteady computations and the rotor heat-transfer measurements was demonstrated. An important result from Sharma's unsteady numerical simulations is that time-averaged loads on airfoil surfaces (and endwalls) are not affected by unsteadiness, even in the presence of temperature streaks. Unsteady computations clearly show that the pressure sides

Thermocouple signals and heater electrical connections from the rotor to the stationary frame of reference were transmitted

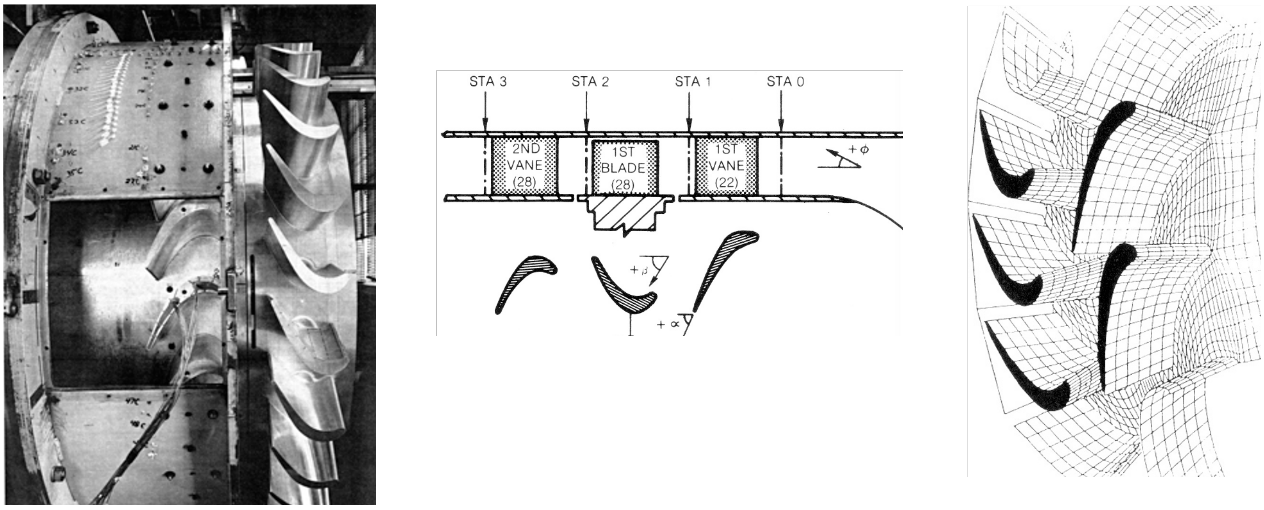


Figure 7. UTRC's Large Scale Rotating Rig (LSRR), Dring and Joslyn [1981]

of turbine airfoils operate at higher temperatures than the suction sides when circumferentially non-uniform temperature profiles are present.

Blair, Wagner and Steuber [1991] developed two interesting liquid crystal techniques to obtain steady-state heat transfer data on the gas side of rotating turbine airfoils in LSSR. In one study, researchers developed a liquid crystal-based method to measure heat-transfer rates on rotating blade surfaces. The second study is on heat transfer in rotating turbine coolant passages. Figure 9 shows the sample liquid crystal temperature patterns on the rotor airfoil at two rotational speeds (N = 407 rpm and 160 rpm).

by using a slip-ring device.

Figure 9 shows the contour maps of Stanton numbers on the suction side and pressure side by using a rectangular projection of the unwrapped airfoil surface. Liquid crystal patterns were used to supplement the thermocouple data in regions with extremely localized effects. A separation bubble from the heat transfer measurements was clearly visualized. A highly three-dimensional pattern near the suction side, especially at the tip, indicates the influence of the tip leakagevortex and secondary flows on the surface heat-transfer distribution. Strong variations in Stanton number

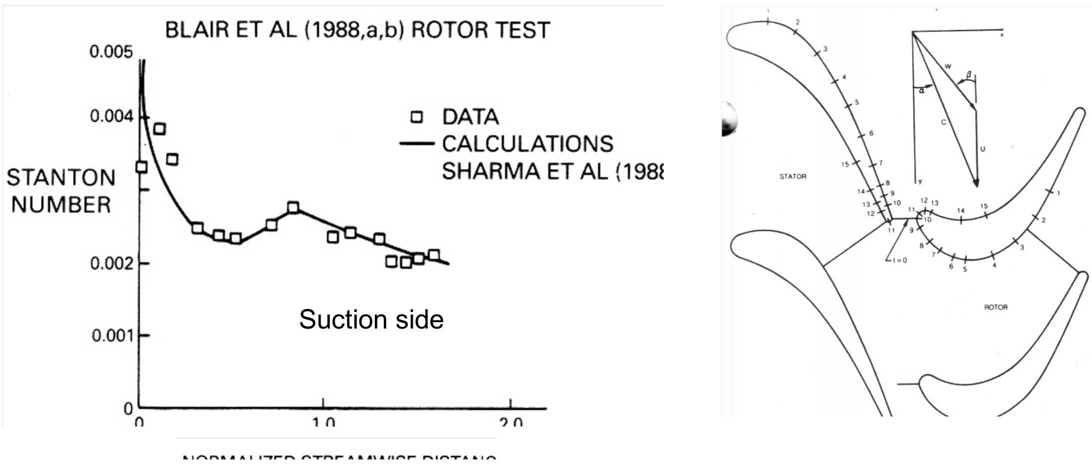


Figure 8. Comparison of LSRR rotor heat transfer data with computations Blair et al. [1988,a,b]

This approach used slightly non-rectangular "constant heat flux surfaces" to measure heat transfer coefficients. The deviations from the constant heat flux value due to slightly non-rectangular heater shapes were computed by analytically evaluating non-uniform current-density functions in the actual heater shapes. The liquid crystal images were obtained in the stationary frame of reference by using a strobe light during the passage of the rotor blades. The rotor was instrumented with arrays of 89, 124, and 102 thermocouples on the pressure side, suction side, and endwall surfaces, respectively.

contours near the suction side of the leading edge can be attributed to horseshoe vortex formation driven by a momentum deficit in the inlet endwall boundary layer.

Turbine Stage in the Oxford Rotor Facility ILPT: Turbine stage in the Oxford rotor facility, as shown in Figure 10, uses an **I**sentropic **L**ight **P**iston compression **T**unnel ILPT to obtain well-simulated gas turbine conditions. Schultz et al. [1977] showed that the temperature differences encountered in gas turbine components could be reduced in light-piston compression-tube experiments. The validity of this approachwas established by Hilditch and Ainsworth [1990].

The turbine stage is a transonic shroudless turbine of 0.5 m

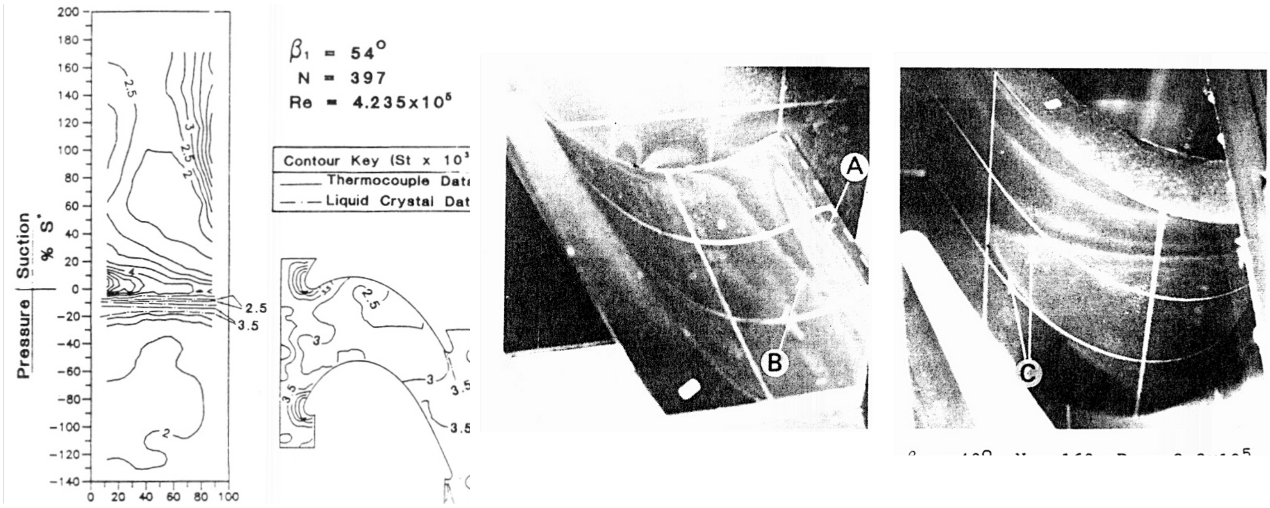
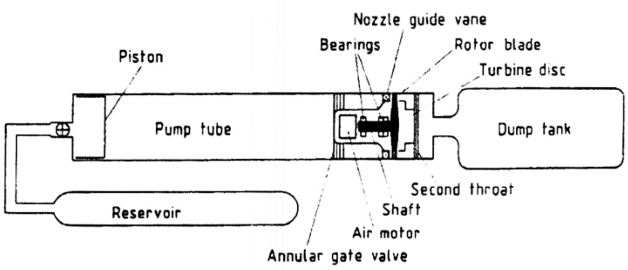


Figure 9. Stanton number measurements on the rotor surfaces of (LSRR) Blair et al. [1991]



Parameter	Symbol	Note	Design Point Value
Rotor Speed	rpm		8343
Specific Speed	$N/\sqrt{T_0}$	-	436 RPM/K$^{1/2}$
Mass Flow Number	$\frac{\dot{m}\sqrt{T_0}}{P_0}$	-	7.04×10^{-4} ms√K
Reynolds Number	Re	Based on NGV exit conditions and axial chord at mid-height	2.7×10^{6}
NGV exit Mach Number	M_{exit}	Isentropic, mid-height	0.946
Rotor relative exit Mach Number	M_r	Isentropic, mid-height	0.959
NGV Axial Chord	c_{NGV}	-	0.0312

Ref: Hilditch [35]

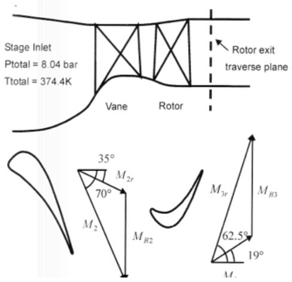


Figure 10. Turbine stage in the Oxford rotor facility ILPT, Ainsworth et al.[1988], Allan et al.[2004]

tip diameter as explained in Ainsworth et al. The specific turbine has 36 NGVs and 60 high-pressure turbine blades. The NGV exit Reynolds number is 2,700,000, and the pressure ratio of the turbine is 3.12. The rotational speed of the rotor is 8343 rpm and the blade was geared toward measuring tip clearance is about 2.25 % of the blade span. Other details of the facility operation and stage schematic are shown in Figure 10.

Figure 11 shows the mid-span thin-film heat-flux sensors and a typical heat-flux distribution from a pressure-surface heat-flux sensor. The oscillations in the heat-transfer signal are due to the piston's finite weight in the compression tube facility. 180 ms of the short-duration test is shown in the figure. Unsteady heat flux signals from the frontal part of the suction side are presented in Figure 11. The large fluctuations beginear the leading edge of the suction side start to diminish in magnitude at about x/S=0.351. High fluctuation levels are usually explained by the passage of the highly turbulent NGV wakes, Allan et al. [2004]. Figure 11 shows a double peak from all sensors between x/S=0.068 and 0.232. The spacing between the peaks varies due to the upstream motion of the shock emanating from the trailing edge of the NGV and the downstream motion of the wake segment. Allan et al. also observed sharp heat flux spikes at x/S=0.479 on the suction side. They attributed the spikes to the presence of reflected trailing edge shocks from the adjacent rotor blade.

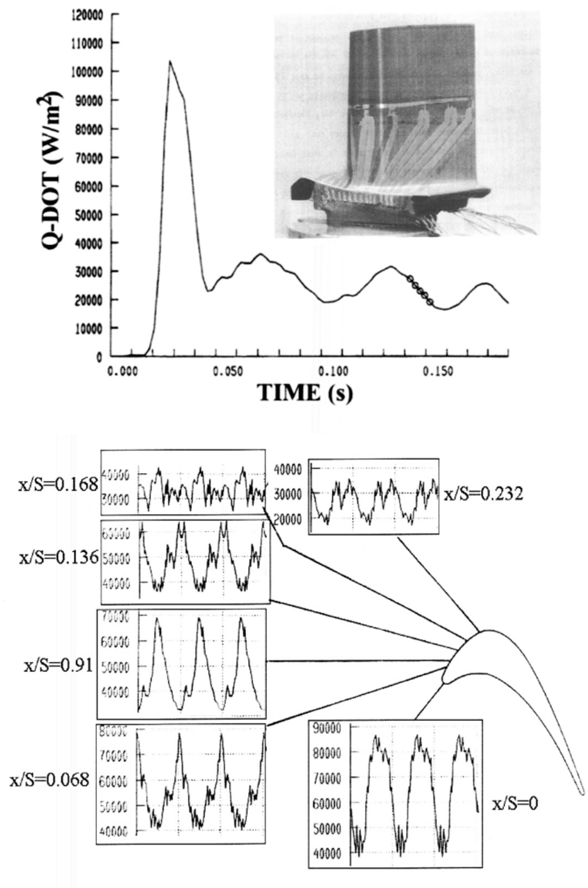


Figure 11. Unsteady heat flux sensors and heat flux measurements in turbine stage in the Oxford rotor facility ILPT, Allan et al., [2004]

The VKI Compression Tube Turbine Test Facility CT3: The VKI research turbine stage shown in Figure 12 is a representative of a wide class of aero-propulsion engines. The stage is driven by a light piston compression tube facility, CT3 that allows the adjustment of the Reynolds number and Mach number, the free-stream to wall and coolant to wall temperature ratios under realistic gas turbine conditions, Sieverding and Arts [1992], Denos [1996] and Sieverding et al. [1998]. Figure 12 shows the original facility configuration, which consists of 43 trailing-edge-cooled guide vanes and a rotor system with 64 blades. The inset shows the modified turbine stage, including a second stator system with 43 vanes. The second stator can be rotated in the azimuthal direction to investigate clocking effects. The tip clearance that has a weak variation from the leading edge to the trailing edge is approximately 0.6 % of the rotor blade height. The operational conditions of the turbine rig are summarized in Figure 13.

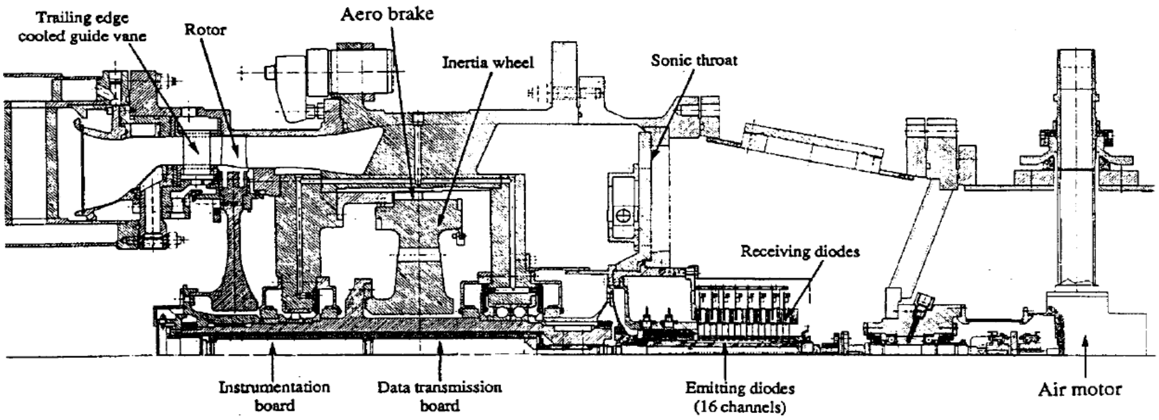


Figure 12. The VKI turbine research rig, Sieverding and Arts [1992], Dénos, R., [1996]

The VKI turbine research rig uses both single-layer deposited thin-film sensors and two-layer gauges to acquire unsteady heat-flux signals. A two-layer "Senflex" thin-film heat-transfer gauge array attached to the suction side of the second stator is shown in Figure 13. A successful implementation of this specific heat flux sensor array is described in detail by measurements performed on the second stator of the VKI turbine. The signals were obtained from two pressure side locations in the first 1/3 chord region. Four different repeats from the two sensors indicate excellent repeatability. The results are ensemble-averaged over three complete rotor revolutions, or 192 rotor-passing events. The phase-locked measurement approach allows the researcher to extract the random, unsteady part of the signal. Quantifying wake passage and potential interactions between the rotor and the stator is essential for interpreting unsteady wall-heating phenomena in transonic turbine stages. Figure 14 also shows the mean wall temperature, wall heat flux, and heat-transfer distribution on the pressure and suction sides of the second stator. The figure shows five runs of the VKI turbine. Each run is obtained from a different initial wall temperature distribution in the rig. Since the turbine flow conditions are accurately

1 & ½ STAGE FLOW CONDITIONS			
P_{01} [bar]	2.23		
T_{01} [K]	480		
P_{01}/P_{03}	2.63		
P_{s4}/P_{03}	0.72		
Rotational Speed [RPM]	6500		
M_{2is}	1.06	$Re_{2,Cs}$ (*)	$1.2x10^6$
$M_{3is,r}$	0.89	$Re_{3r,Cr}$ (*)	$5.2x10^5$
M_{3is}	0.42		
M_{4is}	0.68	$Re_{4,Cs}$ (*)	$4.8x10^5$

* Based on chord & outlet velocity (relative frame for rotor)

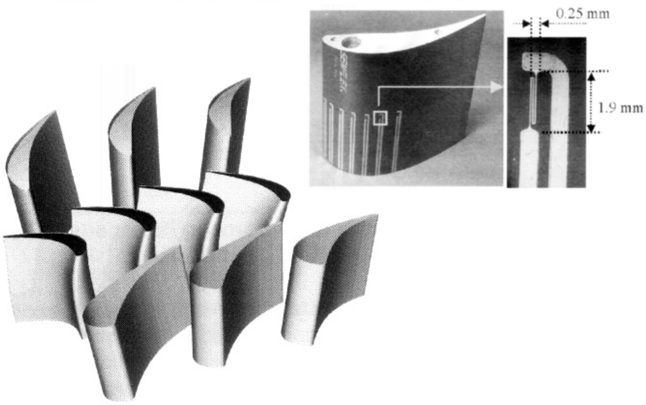


Figure 13. 1 ½ stage VKI turbine research rig operational conditions, Sieverding et al. [1992,1998], and two-layered "Senflex" thin film heat transfer gauge array attached on the suction side of the second stator, Illiopoulou et al. [2004]

Figure 14 presents unsteady Nusselt number traces in function of the rotor position from phase-locked heat transfer

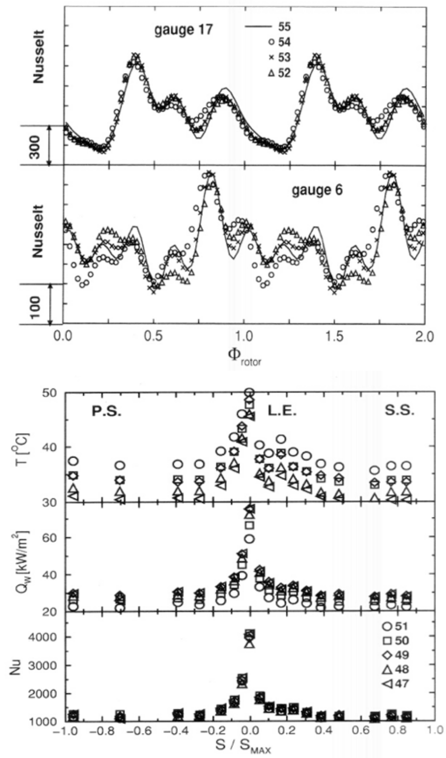


Figure 14. Phase-locked unsteady heat transfer measurements and time-averaged heat transfer on the second stator of the VKI turbine research facility, Illiopoulou et. al (2004)

repeated during each of the five individual experiments, the resulting Nusselt number distributions are almost the same, as shown in Figure 14. A three-dimensional computational simulation of the VKI transonic turbine stage is discussed in Michelassi et al. [2001]. The effect of the hub endwall cavity flow on the flow field of the VKI turbine stage is presented in Paniagua et al.[2004]. The effect of clocking on the aerodynamic and mechanical behaviour is investigated by Gadea et al. [2004].

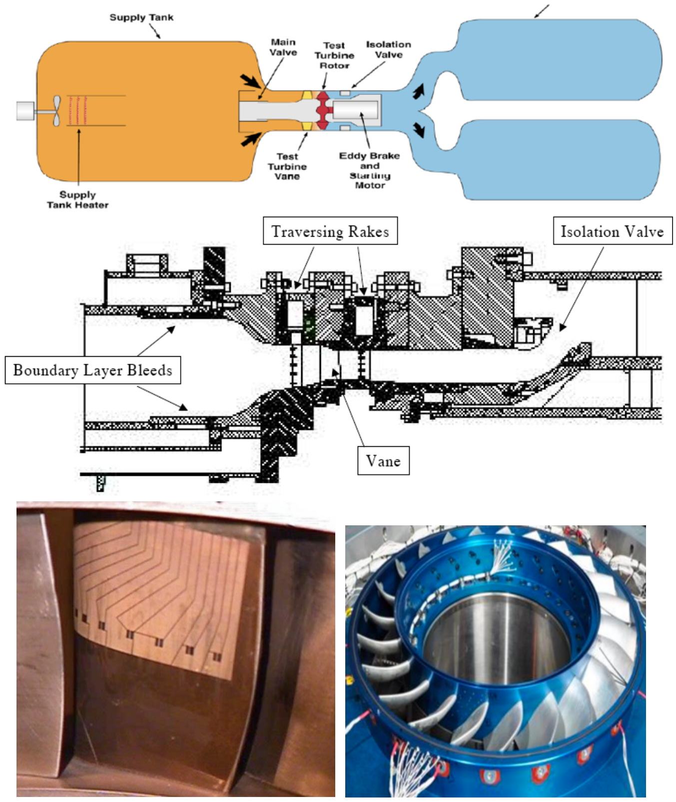


Figure 15. Short duration turbine research facility (TRF) stage and vane suction side heat transfer instrumentation at Wright Patterson Air Force Base, Haldeman et al. (1992)

Short Duration Turbine Research Facility (TRF) at Wright Patterson Air Force Base: Turbine Research Facility (TRF) at the Air Force Research Laboratory (Haldeman et al, 1992) is presented in Figure 15. TRF is a transient "blow down" rig capable of operating for periods of several seconds at the correctly scaled Mach and Reynolds numbers of the turbines of actual engines. Before the start of a TRF test, the entire facility is evacuated, and the fast-acting valve between the supply tank and the test section is closed. Next, the supply tank can be filled with a mixture of N_2 and CO_2 whose composition is set to match the combustion-gas specific heat ratio γ at the engine operating conditions. For most tests, only nitrogen was used, yielding $\gamma = 1.4$ and $Pr = 0.71$. This is reasonably close to the desired Pr of high-temperature air, about 0.68. Figure 15 also shows a vane suction-side surface wrapped with a 50-micron-thick Upilex film sheet, similar to the flexible substrate used at the VKI turbine facility. This sheet was wrapped from the trailing edge to the trailing edge of a single vane, creating no surface discontinuities along the airfoil surface, Polanka and Meininger [2001]. Thirteen 0.04 micron thick platinum sensing elements were sputtered on top of the Upilex flexible substrate prior to installation on the vane suction side. The heat flux data from the gauges were determined by using the resistance change with time to a temperature trace via a calibration for resistance to temperature. The heat flux was then calculated from the time rate of change of the surface temperature, using the thermal properties of the Pyrex or Upilex substrates.

Ohio State University Turbine Research Facility: This rotating turbine heat transfer facility utilized a shock-tunnel to produce a short-duration source of heated and pressurized gas that was passed through the turbine stage, Dunn and Haldeman [2000]. The experimental apparatus shown in Figure 16 consisted of an air-driven shock tube, an expansion nozzle, a dump tank and a test system that housed the turbine stage. The shock tube has a 0.47-m-diameter, 12.2-m-long driver tube and a 0.47-m-diameter, 18.3-m-long driven tube. The driver tube was designed to be long enough so that the wave system reflected from the driver endwall would not prematurely terminate the test. The turbine is housed in a device located in the expansion nozzle of the shock-tunnel

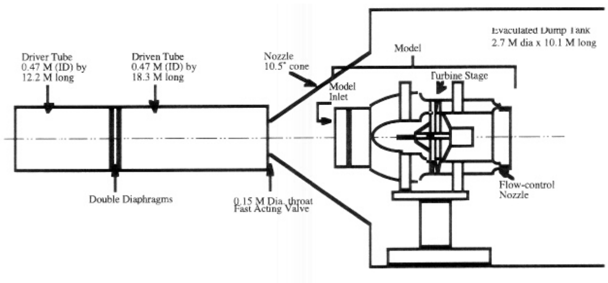


Figure 16. Shock tube driven turbine test facility at the Ohio State University,Dunn and Haldeman (2000)

facility, as illustrated in Figure 16. To start a heat transfer test sequence, the test section is evacuated while the driver and driven tubes are pressurized to predetermined values.

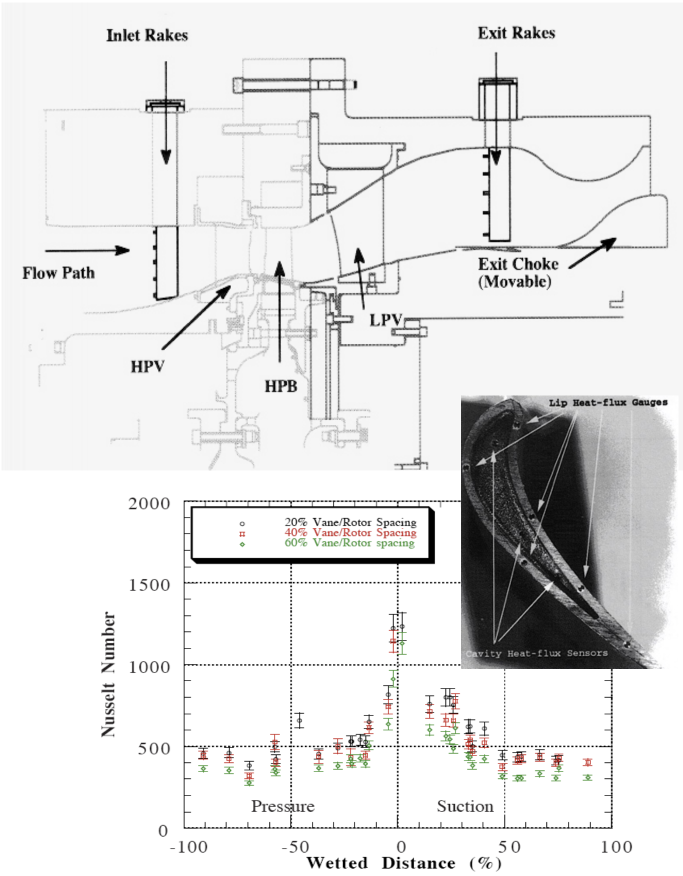


Figure 17. 1 and ½ stage turbine stage at Ohio State University, Haldeman et al. [2004] and Nusselt number distribution, Dunn and Haldeman [2000]

Pressure values are selected to duplicate the design flow conditions. The turbine flow function, wall-to-total temperature ratio, stage pressure ratios, and corrected speed are duplicated. The value of T_o can be set at almost any desired value in the range of 800°R (171 ° C) to 3500°R (1671 ° C). The pressure ratio across the turbine is set by adjusting the throat diameter of a flow-control nozzle near the exit of the turbine housing, Haldeman et al. [2004]. The thin-film heat flux gauges were made of (~100 Å thick) platinum and were hand-painted on an insulating Pyrex 7740 substrate. The response time of these thin films is in the order of 5 x 10^{-8} sec. The turbine stage shown in Figure 17 is a 1 & ½ stage (High-Pressure Turbine Vane-HPTV, High-Pressure Turbine Blade-HPTB, and Low-Pressure Turbine Vane-LPTV) turbine, configured in a manner that is representative of the highly 3D airfoil rows in a modern engine, Haldeman et al. [2004]. The total pressure ratio across the rig exceeds 5. The Ohio State turbine stage shown in Figure 17 has 38 vanes (both HP and LP) and 72 Blades. The exit choke is movable, allowing a variety of pressure ratios. By controlling the pressure ratio across the stage, the inlet pressure, the corrected speed, and a wide variety of engine operating conditions can be replicated. The overall research effort performed in this rig was comprised of the aerodynamics of the turbine stage and unsteady heat transfer measurements performed around the rotor, rotor shroud and vane, Haldeman et al. [2004]. Heat-flux measurements and predictions from the blade tip region of this HP turbine were recently presented by Molter et al. [2006].

Oxford/QinetiQ Light Piston Isentropic Compression Tube Driven Turbine Research Facility: The QinetiQ Isentropic Light Piston Facility (ILPF), as shown in Figure 18, is a short-duration wind tunnel capable of “aero-heat transfer” testing of an engine-size turbine at realistically simulated conditions, Chana et al. [2003]. This facility was originally used to test a single HP turbine stage. The facility was recently

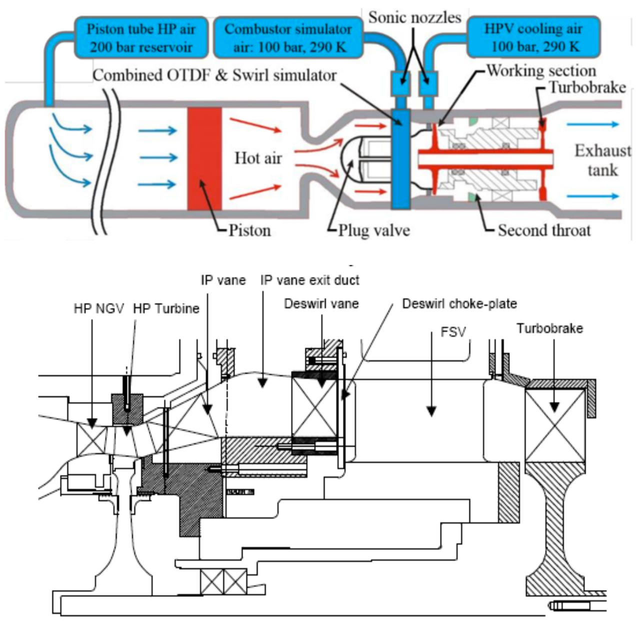


Figure 18. QinetiQ short duration turbine stage, Chana et al. [2003]

expanded to 1.5 turbine stages by installing an IP stator. A unique feature of the facility is the aerodynamic turbo-brake, which is on the same shaft as the turbine and is driven by the turbine exit flow. At the design speed, the turbo-brake power is matched with the turbine, and thus the turbo-brake maintains constant speed during a run, as shown in Figure 18.

Air from a high-pressure reservoir moves a piston down a piston tube, isentropically compressing and heating the working gas (air) inside the tube. When the desired pressure is reached, the compressed air is suddenly released through a fast-acting valve into the working section. Steady–state turbine conditions are achieved for approximately 500 ms. The turbine stage is a modern high-pressure Rolls-Royce aero-engine design with a stage pressure ratio of 3.2 and a nozzle guide vane Reynolds number of 2.54 x 10^6. The stage is unshrouded, and the blade counts for the HP stator, HP rotor, and IP stator are 32, 60, and 26, respectively. Turbine cooling air is delivered from a large reservoir of high-pressure air at ambient temperature. The reservoir pressure is varied to supply the required coolant mass flow, which is metered by an ISO calibrated choked venturi. The facility is also capable of testing with inlet profiles of temperature, inlet swirl and combined temperature and swirl representative of the flow-field exiting modern combustors. A more recent status of the test rig is given by Beard et al. [2019].

Low Speed Large Scale Turbine Rig at Texas A&M University: The three-stage low-speed and large-scale research turbine is the core component of the “Turbomachinery Performance and Flow Research Facility” *TPFL* at Texas A&M University. This facility, designed by Schobeiri [2000], is used to study aerodynamics, efficiency, performance, and heat transfer of high-pressure, intermediate-pressure, and low-pressure turbine components.

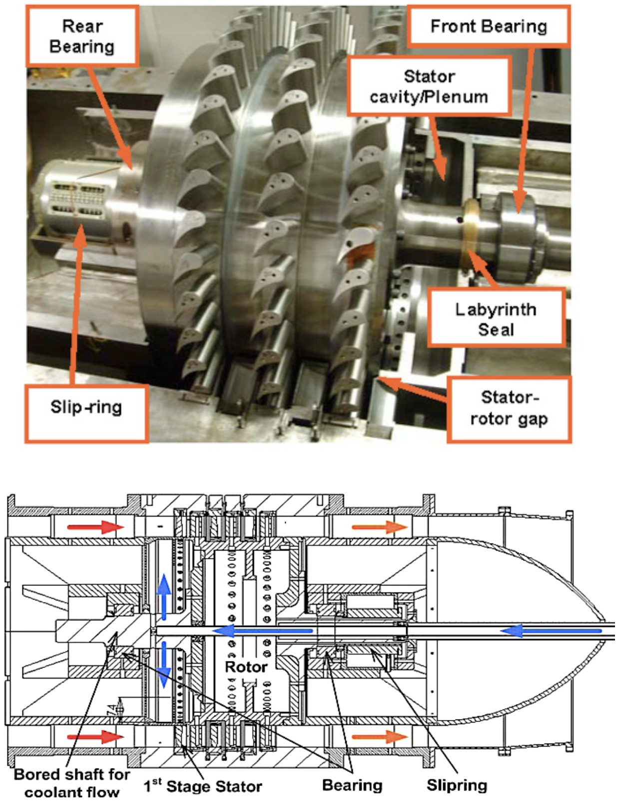
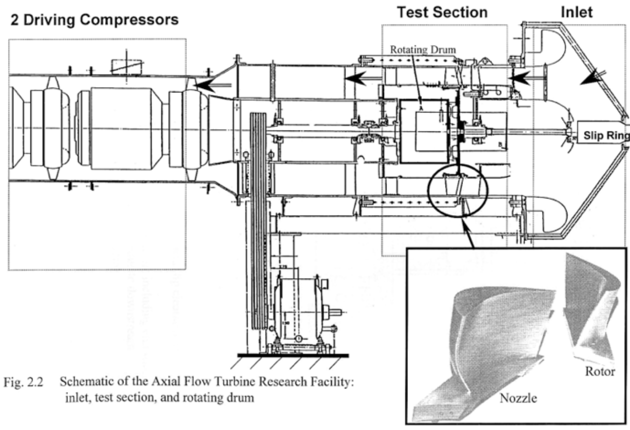


Figure 19 Three-stage, low-speed large-scale turbine facility at Texas A&M University Schobeiri et al. [2000]

The overall layout of the test facility is shown in Figure 19, Suryanarayanan [2006]. The rotational speed of this 0.685 m rotor can be adjusted between 1500 and 2550 rpm. The rotor rows have 46, 40 and 44 blades in a three-stage arrangement, whereas the stator rows have 56, 52 and 48 airfoils. It consists of a 300 HP inverter-controlled electric motor that drives the three-stage compressor in suction mode, with a maximum pressure difference of 55 kPa and a volume flow rate of 4 m^3/s. The turbine inlet has a heater for conditioning the atmospheric conditions. The heater prevents condensation of water from humid air expanding through the turbine during the test. To determine the film-cooling effectiveness under rotating conditions for leading-edge film cooling, the turbine rotor was modified to integrate the coolant loop into the downstream section of the hollow turbine shaft and into the cylindrical hub cavity, as described by Ahn et al. [2005]. This open-to-rig turbine section is among the first rotating facilities where a pressure-sensitive paint-based film-cooling effectiveness measurement technique is frequently and effectively employed.

Axial Flow Turbine Research Facility AFTRF at Pennsylvania State University: The Axial Flow Turbine Research Facility (AFTRF), at the Turbomachinery Aero-Heat Transfer Laboratory, is shown in Figure 20. A recent description of the operational characteristics of this rotating rig, designed primarily for aerodynamic and heat-transfer investigations, is available in Camci [2004].

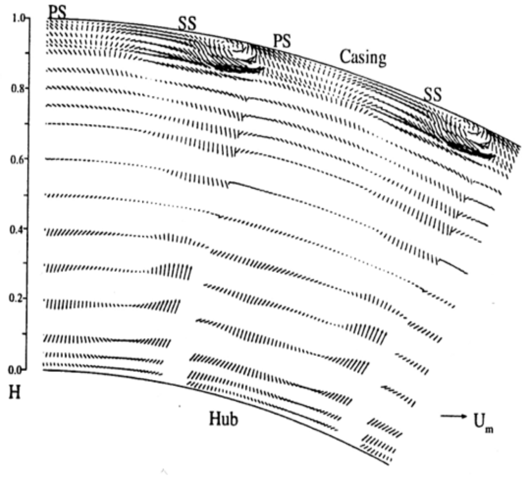


Inlet Total Temperature: T_{o1} (K)	289
Inlet Total Pressure: P_{o1} (kPa)	101.36
Mass Flow Rate: Q (kg/sec)	11.05
Rotational Speed: N (rpm)	1300
Total Pressure Ratio: P_{o1}/P_{o3}	1.077
Total Temperature Ratio: T_{o3}/T_{o1}	0.981
Pressure Drop: P_{o1}-P_{o3} (mm Hg)	56.04
Power: P (kW)	60.6

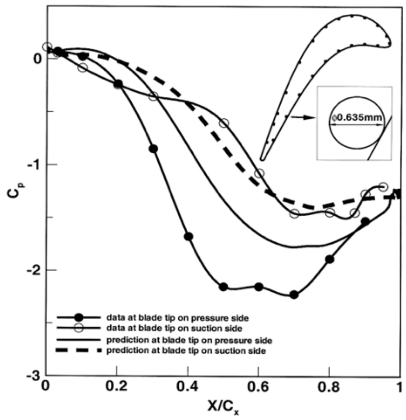
Rotor hub-tip ratio	0.7269
Blade Tip Radius; R_{tip} (m)	0.4582
Blade Height; h (m)	0.1229
Relative Mach Number	0.24
Number of Blades	29
Axial Tip Chord; (m)	0.085
Spacing; (m)	0.1028
Turning Angle; Tip / Hub	95.42° / 125.69°
Nominal Tip Clearance; (mm)	0.9
Reaction, Hub / Tip	0.197 / 0.519
Reynolds Number ($\div 10^5$) inlet / exit	(2.5~4.5) / (5~7)

Figure 20. Axial Flow Turbine Research Facility AFTRF at Pennsylvania State University,Turbomachinery Aero-Heat Transfer Laboratory, Lakshminarayana et al. [1996] and Camci [2004]

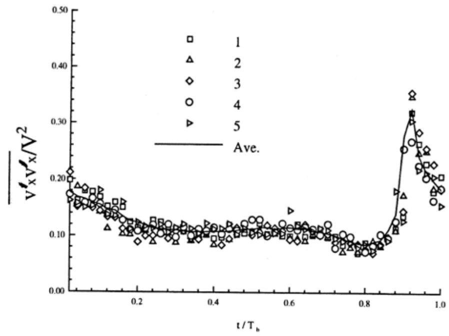
The research facility is a large-scale, low-speed, cold-flow turbine stage depicting many characteristics of modern high-pressure turbine stages. Airflow through the facility is generated by a four-stage axial fan located downstream of the turbine. The turbine rig has a removable casing segment for measurement convenience, especially for casing and tip-related aero-thermal studies. The relevant design performance data includes reaction at the blade hub and tip, Reynolds number at rotor exit and a few blade parameters. are also listed in Figure 20. Measured/design values of rotor inlet flow conditions, including radial, axial, and tangential components, and data acquisition details of the turbine rig are explained in detail by Camci [2004], Rao et al. [2004], and Lakshminarayana et al. [1996]. Instruments used for monitoring the performance parameters of AFTRF consist of total pressure probes, Kiel probes, pitot-static probes,



Phase locked LDA measurements at the rotor exit



Static pressure near the blade tip



Turbulence intensity behind the rotor row

Figure 21 Rotor exit flow field, rotor tip static pressure and rotor exit axial turbulence intensity in AFTRF, Turbomachinery Aero-Heat Transfer Laboratory, Camci [2004] and Lakshminarayana et al. [1996]

thermocouples, and a precision in-line torquemeter. The turbine rotational speed, which is adjustable to 1 rpm, is kept constant at approximately 1300 rpm by an eddy current brake.

Figure 21 shows AFTRF rotor-related aerodynamic details that can be highly influential in the determination of final wall heating patterns in axial turbines. A set of phase-locked LDA measurements clearly shows the extent of tip vortical flow near the tip, wake pattern and rotor secondary flow structure. The middle figure presents the wall static pressure measurements performed near the blade tip in the rotating frame of reference. Although the measured suction-side pressure coefficient agrees well with the RANS computations, the static pressure measurements near the pressure-side corner, where tip leakage is significant, show a discrepancy relative to the computational results. This may be attributed to the highly strained, separated relative flow pattern in this region, where static holes may experience leakage flow partially impinging on them. Finally, the rotor exit mid-span circumferential turbulence intensity shows that high levels of free-stream turbulence, ranging from 10% to 35%, are typical in the wake near the rotor exit plane.

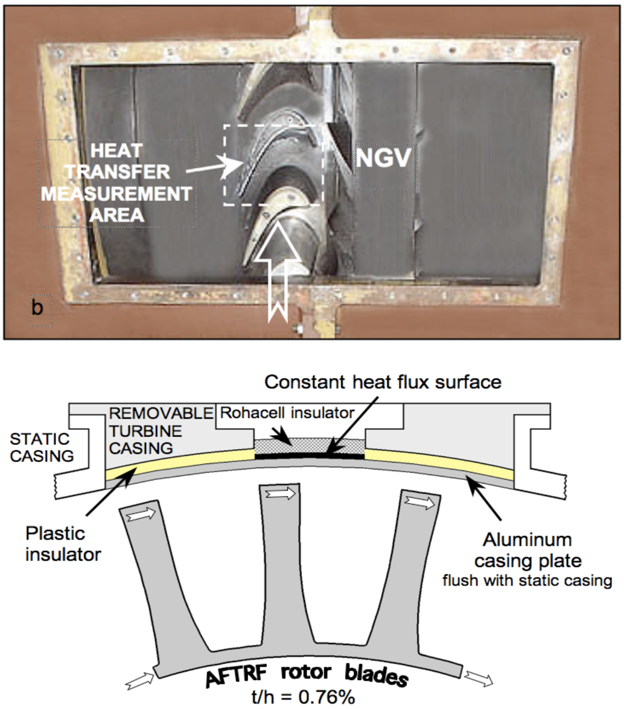


Figure 22. Heat transfer measurements using a constant heat transfer surface near the casing in AFTRF, Turbomachinery Aero-Heat Transfer Laboratory, Gumusel and Camci [2010]

The most recent aero-heat transfer research at AFTRF focuses on the tip and casing regions. Figure 22 shows the casing heat-transfer measurement system mounted on the flow side of the "turbine access window" and flush-mounted in the research rig. This system was developed for measuring convective heat transfer coefficients on various "patterned" casing surfaces covering the rotating blade tip area. The flow side of the casing plate is flush-mounted with the rest of the AFTRF casing. The axial distribution of the heat transfer coefficient is measured with an estimated uncertainty level between 5 % and 8 % of the convective heat transfer coefficient *h*, Gumusel and Camci [2009]. The heat transfer results shown in Figure 23 are obtained from a 3D heat-loss analysis to reduce measurement uncertainties. A 3D heat-conduction analysis of the casing window and the casing plate is conducted simultaneously. The reference free-stream temperature for this turbine convective heat-transfer problem is the "adiabatic wall temperature," $T_f = T_{aw}$. Due to energy extraction from the working fluid in a turbine, the total temperature decreases continuously along the axial direction. Therefore, an accurate method of obtaining the true reference temperature (adiabatic wall temperature) is needed. Figure 23 shows the results of recent experiments -heat-in which the power setting of the constant heat flux surface is continuously varied to obtain the local "adiabatic wall temperature" at a selected axial measurement position. The dramatic influence of working with the correct reference temperature in an axial turbine rotor environment is apparent in the middle part of Figure 23. The adiabatic wall temperature is a strong function of the axial position in the rotor frame of reference, and it needs to be measured accurately in order to keep heat transfer coefficient uncertainties in the 5-8 % error band, Gumusel and Camci [2009].

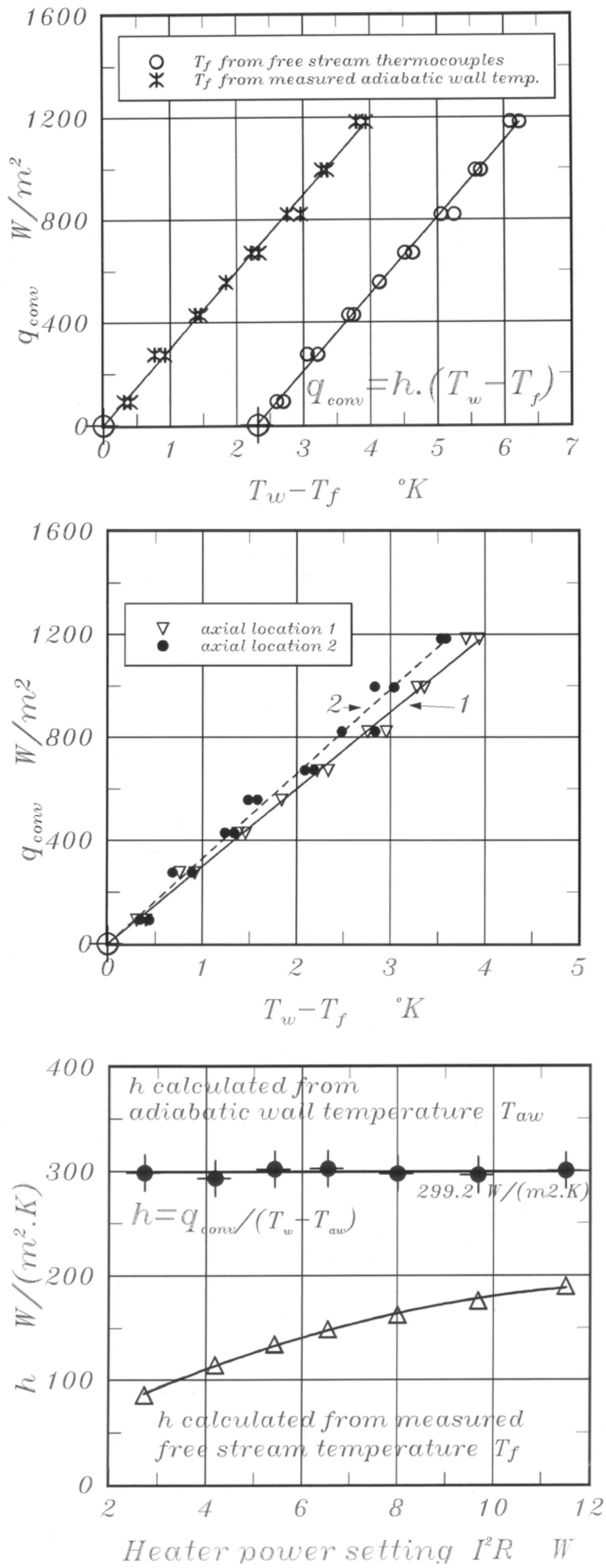


Figure 23. Determination of the adiabatic wall temperature and heat transfer coefficient on the casing surface of AFTRF, Gumusel and Camci (2010)

Axial Turbine Rearch Facility "LISA" at ETH Zurich: The facility shown in Figure 24 is a modular, moderate-speed, low-temperature test rig that supports both 2-stage and 1.5-stage turbine configurations. A detailed description of the 2-stage design, including the modifications for the 1 ½ stage design, is provided by Behr et al. [2007]. The facility operates as a quasi-closed air loop, with the atmosphere open downstream of the turbine section. A 750 kW radial compressor upstream of the turbine drives the flow and provides a maximum pressure ratio of 1.5. At constant rotational speed, the compressor

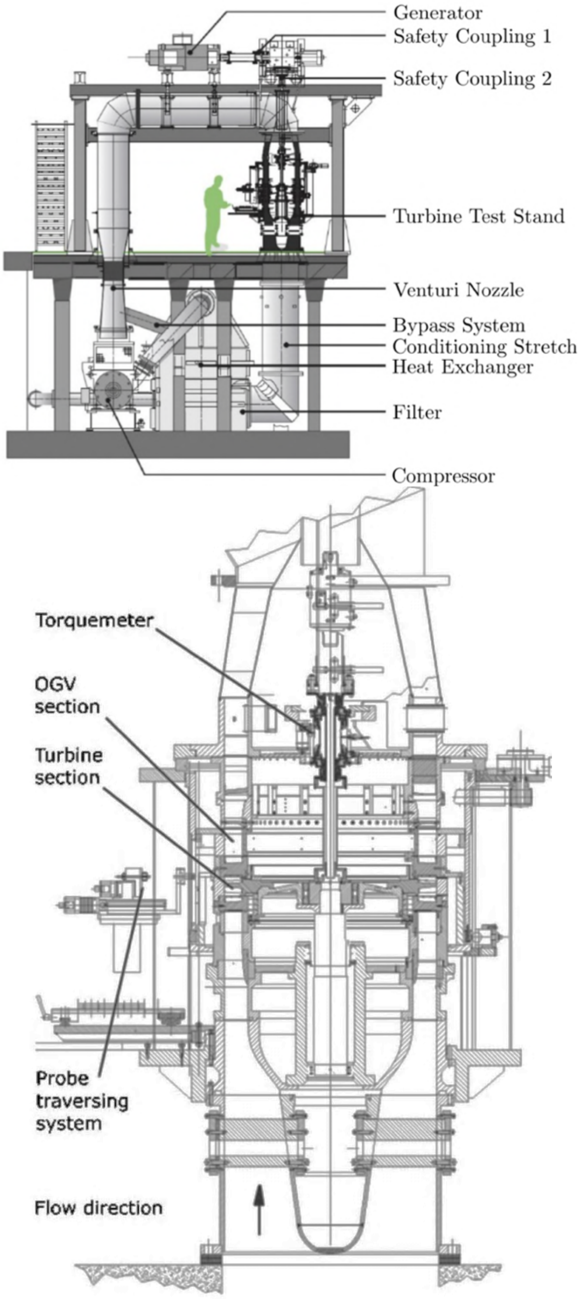


Figure 24. Schematic view of the “LISA” research axial turbine at ETH Zurich, Behr et al. [2007].

pressure ratio and mass flow are controlled using variable inlet guide vanes. The main mass flow is measured upstream of the compressor using a calibrated venturi nozzle. Turbine inlet temperature is regulated by a two-stage water-to-air heat exchanger with an accuracy of ±0.4 ºK. Total inlet pressure is measured by Pitot probes mounted in thin struts located approximately three axial chords upstream of the test section. The turbine shaft is connected via an angular reduction gearbox to a DC generator, which controls rotational speed with an accuracy of ±0.02% (±0.5 rpm). The generated power is fed back into the electrical grid.

Next Generation Turbine Test Facility (NG-Turb) at the German Aerospace Center (DLR) in Göttingen: NG-Turb is a continuously operating closed-circuit facility for aerodynamic turbine investigations, allowing independent variation of engine-relevant Mach and Reynolds numbers. The facility is described in detail by Rehder et. al [2017]. Dry air is used as the working fluid and is driven by a four-stage radial gear compressor, which provides a high pressure ratio and a wide range of inlet volumetric flow rates. In its first stage, the NG-Turb test section enables investigations of single-shaft turbines up to 2.5 stages. In a later expansion stage, a second independent shaft system is installed to enable experiments on combined high- and low-pressure turbine configurations, including counter-rotating turbines. Secondary air for cooling studies can be supplied by auxiliary screw compressors. The turbine mass flow is measured redundantly using calibrated Venturi nozzles upstream and downstream of the test section, with an uncertainty of approximately ±0.3%. The test section enables modular preassembly of turbine rigs, including the use of combustor simulators and good spatial and optical access for installing extensive measurement techniques. The main flow from the compressor rig is divided into a hot and a cold line (via a water cooler). The two streams are then mixed in the pipe system far upstream of the turbine test section, ensuring that mixing is complete before entering the turbine rig. By controlling the mass flow ratio of the hot and cold stream with variable flaps, the inlet temperatures to any installed turbine rig can be adjusted with an accuracy of ±1ºK. To better control inlet temperatures, it is possible to install an additional heater in the hot line. All hot parts of the flow circuit are insulated with 200mm thick mineral wool mats. Test section exit temperatures depend on the turbine's operating points. To avoid compressor inlet temperatures exceeding 300ºK (the design limit), a second water cooler is installed downstream of the test section.

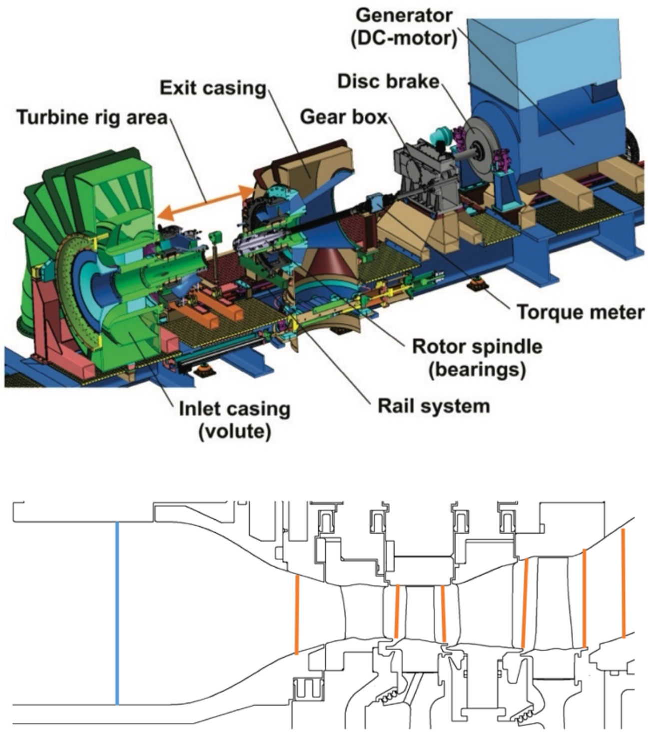


Figure 25. (NG-Turb) at the German Aerospace Center (DLR) in Göttingen, Rehder et.al [2017]

University of Notre Dame’s Transonic Research Turbine (TRT) facility: The TRT facility is a modular test rig designed for continuous, high-speed operation of low- and high-pressure axial turbines, as schematically shown in Fig. BBB. It operates as a semi-closed loop in which “cold” (<100°F) ambient air is mixed with “hot” (~300°F) exhaust air using a lobed mixer upstream of the turbine inlet. The turbine inlet temperature is controlled by two electromechanical butterfly valves that regulate the hot-to-cold air mixing ratio. As the flow expands through the turbine stage, its total pressure decreases before it is drawn into a centrifugal compressor driven by a 500 hp AC electric motor operating up to 5000 rpm. The motor shaft is mechanically connected to both the turbine and the compressor via a gearbox. The compressor is an off-the-shelf unit selected to provide mass flow rates and pressure ratios representative of modern axial turbine designs. It is equipped with variable inlet guide vanes (IGV) and diffuser/outlet guide vanes (DGV), allowing the pressure ratio to be adjusted at constant rotational speed. Downstream of the compressor, the flow is pressurized above atmospheric conditions and exhausted to the ambient or recycled. A detailed description of the TRT facility is presented in Kelly et. al [2017].

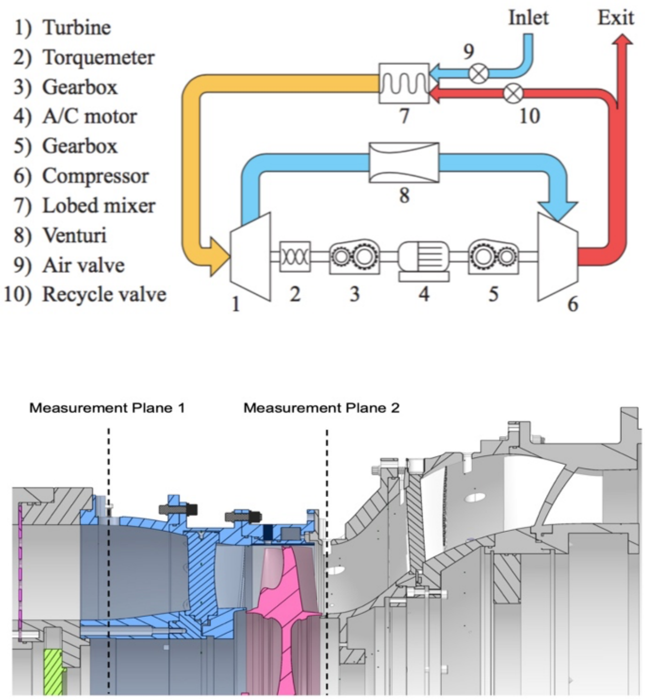


Figure 26. University of Notre Dame’s Transonic Research Turbine (TRT) facility

Steady Thermal Aero Research Turbine (START) at Penn State: A rendering of the Steady Thermal Aero Research Turbine (START) facility at Pennsylvania State University is shown in Figure 27. START is a new gas turbine research facility featuring several unique features that enable experiments to develop and improve sealing and cooling technologies at a reasonable cost. These experiments will include engine-representative rotating turbine hardware operating in a continuous, steady-state, high-pressure flow environment. The facility includes a 1.5-stage turbine that simulates the aerodynamic flow and thermal-field interactions in an engine between the hot mainstream gas path and the secondary air flows, under relevant corrected operating conditions and scaling parameters. The START Lab houses a continuous-duration, open-loop facility supporting a single-stage test turbine. Its steady-state operation is enabled by two 1500 hp compressors, which provide a total continuous mass flow rate of 11.4 kg/s (25 lbm/s) at 480 kPa (70 psia) and a nominal temperature of 380 °K (230°F). The air is further heated by a natural gas burner, raising the temperature to a maximum of 675 °K (750°F). Cooling air is bled from the process stream upstream of the burner and cooled to 273 °K (32°F) in a shell-and-tube heat exchanger before being supplied to three coolant flows.

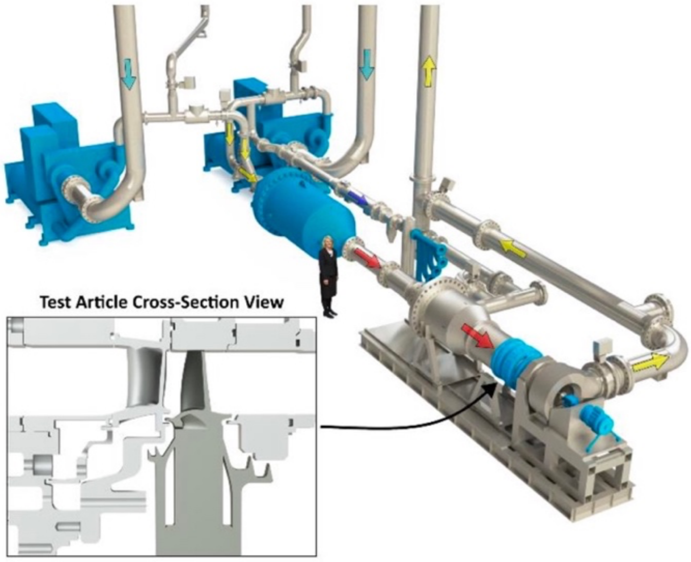


Figure 27. The rendering of the START Facility at Penn State, Barringer et. Al [2014]

The test section consists of a one-and-a-half-stage assembly that includes engine hardware and several additively manufactured vane doublets with integrated static-pressure taps. The rotor assembly is supported by a magnetic levitation bearing system, and its rotational speed can be controlled to ±10 RPM at up to 11,000 RPM using a water brake. A comprehensive report on the facility's development is provided by Barringer et al. [2014], and updates are documented by Berdanier et al. [2019].

Multi-Purpose High-Speed Rotating Rig At The Whittle Laboratory: The test rig can operate in multiple modes, enabling evaluation of axial compressors, turbines, and fans. It may be powered by an electric motor or driven by incoming airflow, in which case, the motor operates as a generator. The primary purpose of the rig is to enable rapid testing of single stages across a wide range of operating conditions, with blade relative Mach numbers between 0.5 and 1.3 and Reynolds numbers (based on chord) ranging from 80,000 to 2,000,000. A rendering of the rig is shown in Figure 28. The facility is a pressurized, variable-density test rig housed within a closed-loop system and installed at the National Centre for Propulsion and Power (NCPP) at the Whittle Laboratory in Cambridge. Its 1.2 MW driveline can support any single-stage experiment, including high- or low-hub-to-tip-ratio compressors, fans, or turbines. The rig enables independent control of Mach and Reynolds numbers, and the pressurized vessel contains a range of fixed and movable sensors for measuring performance and flow characteristics. Using a pressurized closed-loop system offers two key advantages. First, it reduces mechanical stresses on the test hardware. As the pressure vessel shown in Figure 28 carries the pressurization load, the experimental casings inside do not require pressurization certification, which simplifies both mechanical design and manufacturing. Second, the system allows Mach and Reynolds numbers to be controlled independently during testing. A vacuum pump can evacuate the loop to 0.3 Bar, while a screw compressor can increase the pressure up to 8 Bar. Since understanding flow mechanisms for a given technology often depends more on observed trends than on absolute non-dimensional values, this configuration allows sweeps or matrices of Mach and Reynolds numbers to be completed within a single test. This significantly increases the value of the data collected while using the same hardware. Other details of this high-speed rig is given in Harman et. al [2024].

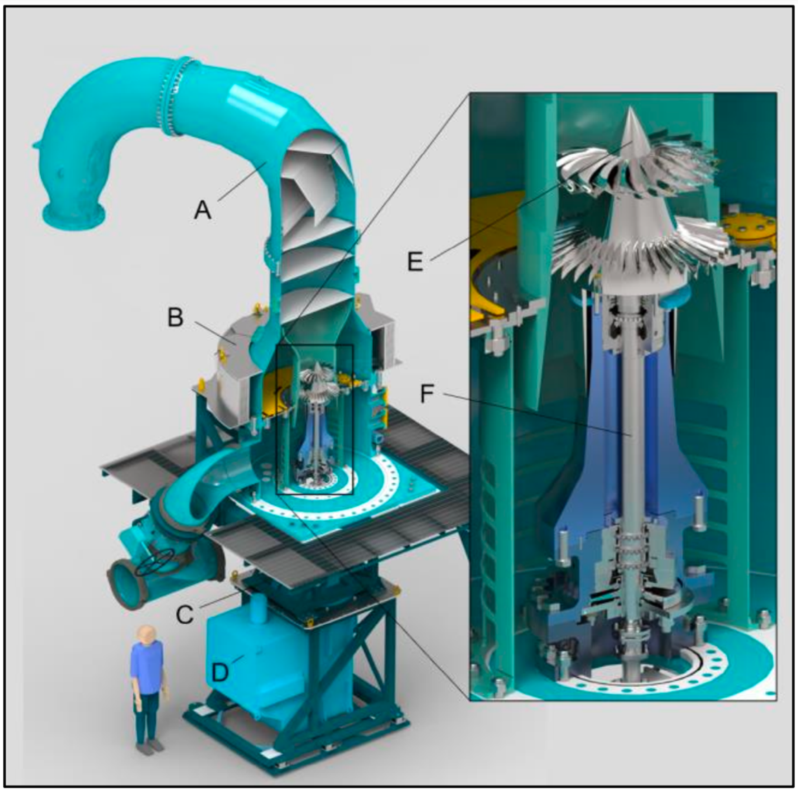


Figure 28. Multi-Purpose High-Speed Rotating Rig, (A) Pressure vessel, (B) Containment ring, (C) Gearbox, (D) Motor, (E) Test articles, (F) High-speed shaft, Harman et.al [2024]

The Small Turbine Aerothermal Rotating Rig (STARR) at Purdue University: Figure 29 presents the schematic of the pressure-driven STARR facility located in the Purdue Experimental Turbine Aerothermal Laboratory (PETAL). The facility produces aero-thermal, engine-representative test conditions using a high-pressure dry air reservoir at 138×10^5 Pa. The reservoir supplies three separate air lines that feed the wind tunnel and the test article. One line is heated using a heat exchanger in a natural gas heater, and the heated flow is then mixed with unheated air from a second line to achieve the desired inlet temperature. The mixed flow rate is measured with a critical Venturi nozzle and directed by fast-actuation butterfly valves either to ambient or into the test section. Upstream of the turbine rig, a settling chamber rated for $P_{max} = 8$ bar and $T_{max} = 700$ °K ensures uniform turbine inlet conditions Paniagua et al. [2019]. The reservoir also supplies secondary air systems, including purge and cooling flows, cavity pressurization, and lubrication system energization.

Downstream of the test section, a sonic valve controls the Mach number distribution across the turbine, while discharge to a vacuum tank enables adjustment of the Reynolds number distribution. The test section of the Small Turbine Aerothermal Rotating Rig (STARR), as also described in Cuadrado et al. [2019], is shown in Figure 30, provides an unscaled, engine-representative geometry of the small-core, high-pressure, high-speed two-stage turbines. The STARR rig reproduces engine-relevant blade loading, inter-disk-cavity purge flows, and secondary-flow characteristics. The turbine shaft is directly coupled to a 780 kW high-speed synchronous electric motor, enabling operation up to 20,000 rpm in both motoring and loaded configurations. The blades analyzed in this work feature squealer tips with a continuous rim. The stationary surface adjacent to the rotating blade tips is covered with an abradable coating applied to the surrounding casing disks.

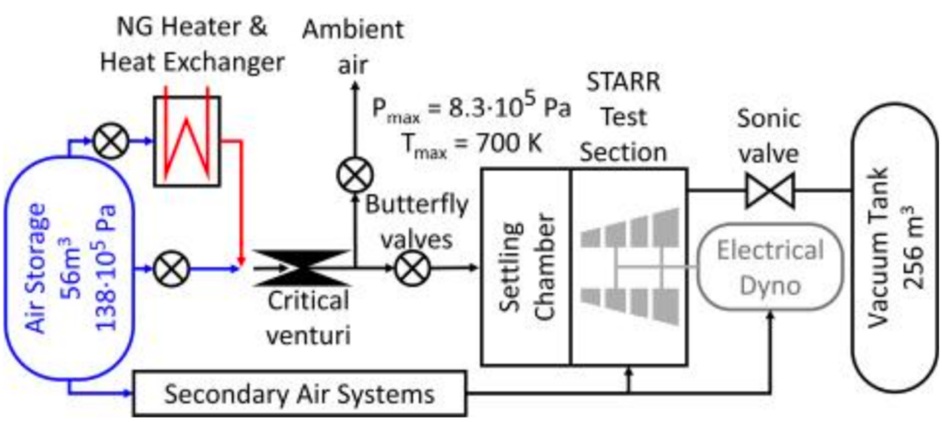


Figure 29. STARR turbine test facility at Purdue, Paniagua et al. [2019]

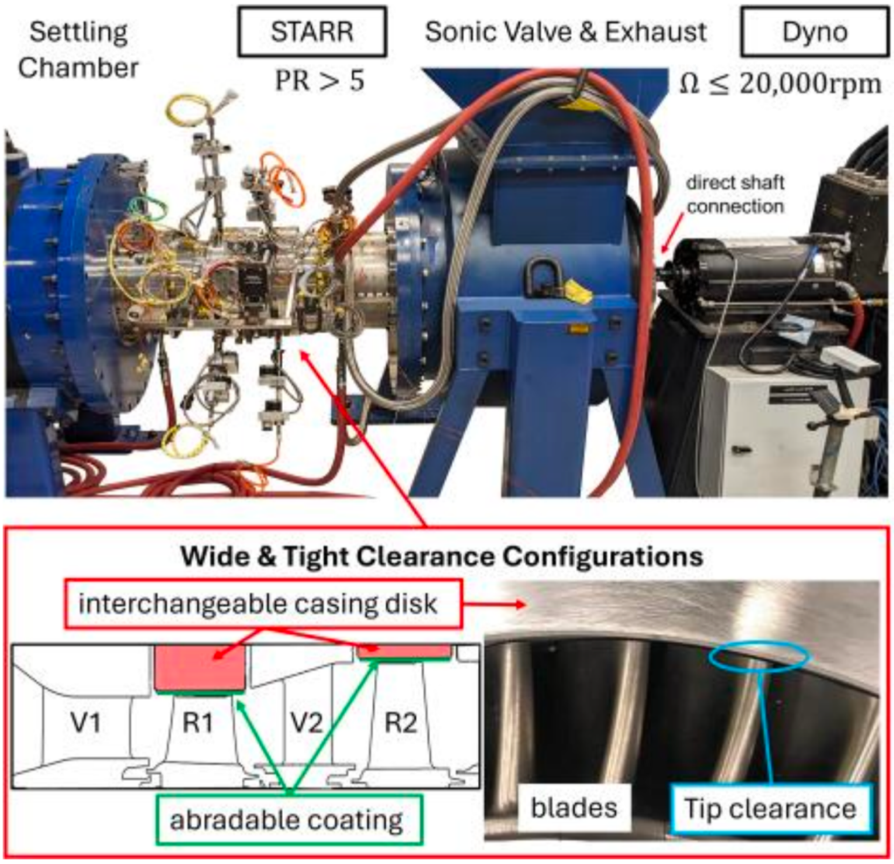


Figure 30. STARR turbine test facility with interchangeable casing disks and abradable coating conforming to tip clearance configurations, Paniagua et. al [2019]

7. Conclusions

This paper presents experimental aero-heat-transfer studies conducted in rotating turbine research facilities. Because the full-scale operational conditions of modern gas turbines require temperatures above 3600°F and pressure ratios of 20-50, experimental forced-convection heat-transfer research on the gas side of a rotating turbine is technically challenging. This paper provides a brief review of turbine heat transfer research in various rotor-containing facilities, including short-duration blowdown and large-scale/low-speed turbine systems. Since the final status of any forced-convection heat-transfer problem is closely related to the detailed structure of momentum transfer in highly unsteady, rotating, 3D, and turbulent viscous-flow environments, emphasis was also placed on supporting turbine aerodynamics research.

After a discussion of the proper scaling of the gas turbine convective heat transfer problem, the rotating facilities used to produce heat transfer data are selected from the most recent open literature publication. The rotating turbine research rigs performing "only" stage aerodynamics/performance measurements, disk cavity flow experiments, rotating coolant passage experiments, and turbine seal flow experiments are excluded from this heat transfer-focused publication.

The final form of the energy equation suggests that one needs to simulate the Eckert number by operating at a realistic Mach number M, the free-stream-to-wall temperature ratio T/T_w, and the specific heat ratio. This condition is in addition to Re and Pr, "for proper simulation ability" in a forced convection heat transfer problem. Simulation of a gas turbine environment also needs to include a proper mass-flow function, corrected speed, and a rotation number, also called the Rossby number. The last set of requirements directly results from a dimensional analysis of the stagnation enthalpy change, power output, and efficiency of a turbomachinery system.

The scaling issues considered so far have assumed a laminar-flow environment in a momentum and convective heat-exchange problem. In a modern gas turbine, the combustor exit flow exhibits significant turbulence and free-stream unsteadiness. The turbulence intensity and length scale of turbulence are the two parameters that should be considered very carefully. The rotor blades of a gas turbine continuously pass through the highly turbulent, low-axial-momentum wakes of nozzle guide vanes, generating a highly unsteady flow in the rotor passages.

In nozzle guide vanes with a transonic/supersonic exit flow, the rotor blades may also chop through shock waves originating from the NGV trailing edges. In the relative flow environment of the rotor exposed to hot gases, the blade surfaces experience the periodic passage of NGV wakes, shock waves, and high-momentum fluid originating from the central sections of the upstream passage. The wall shear stress generation on gas turbine surfaces and heat transfer to these surfaces are highly three-dimensional and unsteady. When the gas turbine is film-cooled, the typical additional scaling parameters are the blowing rate, which is the coolant-to-free-stream mass flux ratio $M = \rho_c U_c/\rho_\infty U_\infty$ and the coolant-to-free-stream temperature ratio, T_c/T_o. The ratio of the coolant to free-stream momentum flux rate is also a popular cooling performance parameter $I=\rho_c U_c^2/\rho_\infty U_\infty^2$. A mass flow rate ratio of coolant to free-stream for a stage or the entire turbine may also be used as a cooling performance indicator.

Acknowledgements

This paper contributes to the commemorative issue of **ARI**, the archival journal of Istanbul Technical University, published in honor of the remarkable academic and scientific legacy of the late Prof. Dr. Taner Derbentli of the İTÜ, Faculty of Mechanical Engineering. He served as my academic advisor and thesis supervisor from 1975 to 1979. I am his first thesis student. Through his deep commitment to education and research, he profoundly influenced the lives of countless students, including my own. He is dearly missed. Rest in peace, **Taner ağabey**—we will always remember you with love and gratitude.

The author also gratefully acknowledges the continued support of the Department of Aerospace Engineering at the Pennsylvania State University throughout his academic career from 1986 to 2026. He is deeply indebted to the many undergraduate and graduate thesis students, faculty colleagues, and the dedicated technical and administrative staff, whose enthusiasm and contributions have enriched the work of the Aero-Heat Transfer Laboratory.

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